RAJALAKSHMI ENGINEERING COLLEGE
RAJALAKSHMI NAGAR, THANDALAM – 602 105



CS23334 – Fundamentals of Data Science Lab

Laboratory Record Note Book

Name : ....M. Mugesh.....

Year / Branch / Section : ...II - CSE - "B"....

University Register No. : ...2116250701454.....

College Roll No. : .....250701454.....

Semester : .....III.....

Academic Year : .....2026.....

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CS23334 – Fundamentals of Data Science Lab

NAME	M. Mugesh
ROLL NO	2507014S4
YEAR	II Year
SEC	B

RAJALAKSHMI ENGINEERING COLLEGE
An Autonomous Institution

BONAFIDE CERTIFICATE

Name: M. Mugesh

Academic Year: ..... **Semester:** III **Branch:** CSE

Register No. 2116250701454

Certified that this is the bonafide record of work done by the above student in the Fundamentals of Data Science Lab ***Laboratory during the academic year 2026 – 2027.***

Karthi

Signature of Faculty in-charge

Submitted for the Practical Examination held on .....

Internal Examiner

External Examiner

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Exp No:1.a

Analyze the trend of data science job postings over the last decade

Aim:

To analyze the trend of data science job postings over the last

decade

Description:

Use web scraping (e.g., BeautifulSoup) or APIs (e.g., LinkedIn API) to gather

data on the number of data science job postings each year. Use pandas for data manipulation and matplotlib/seaborn for visualization.

Code:

```
import pandas as pd

import matplotlib.pyplot as plt

data = {'Year': list(range(2010, 2021)),

'Job Postings': [150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200]}


df = pd.DataFrame(data)

plt.plot(df['Year'], df['Job Postings'], marker='o')

plt.title('Trend of Data Science Job Postings')

plt.xlabel('Year')

plt.ylabel('Number ofJob Postings')

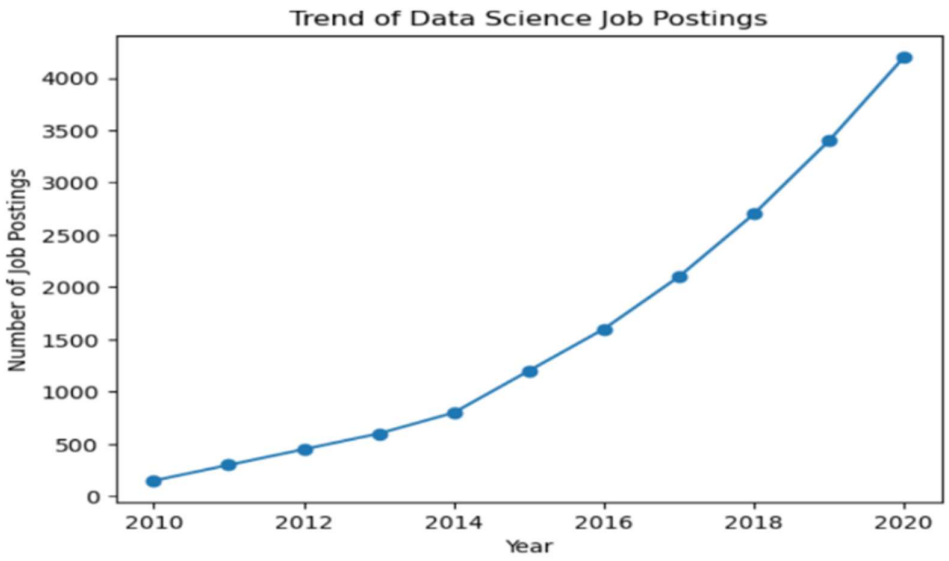
plt.show()
```

Sample Data Input:

Year =2010, 2021

Job Postings=150, 300, 450, 600, 800, 1200, 1600, 2100, 2700, 3400, 4200

Sample Output:



RESULT :

Thus the trend of data science job postings over the last is analysed.

Exp No:1.b

Analyze and visualize the distribution of various data science roles (Data Analyst, Data Engineer, Data Scientist, etc.) from a dataset.

Description:

Use a dataset of job postings and categorize them into different roles. Visualize the distribution using pie charts or bar plots.

Code :

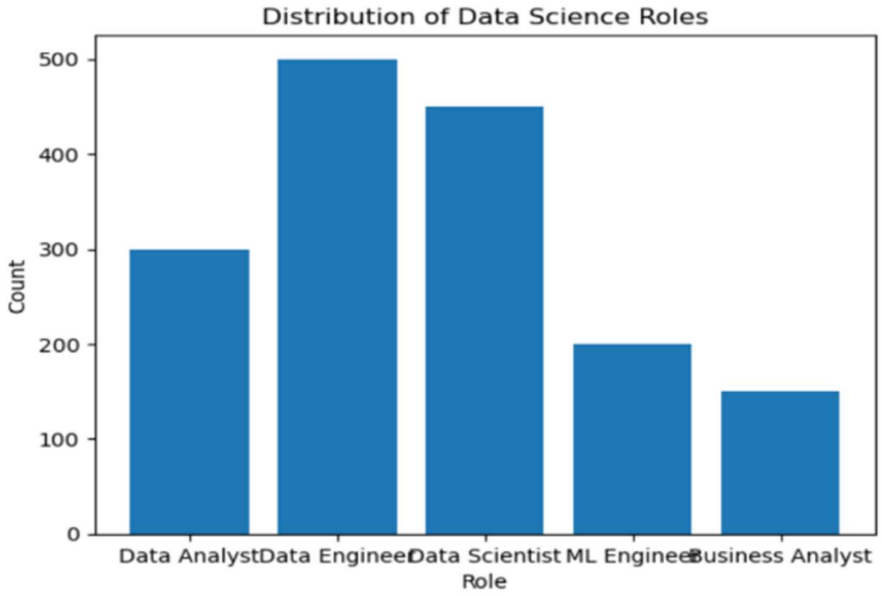
```
roles = ['Data Analyst', 'Data Engineer', 'Data Scientist', 'ML Engineer',
'Business Analyst']
counts = [300, 500, 450, 200, 150]
plt.bar(roles, counts)
plt.title('Distribution of Data Science Roles')
plt.xlabel('Role')
plt.ylabel('Count')
plt.show()
```

Sample Data Input:

roles = Data Analyst, Data Engineer, Data Scientist, ML Engineer, Business Analyst.

counts = 300, 500, 450, 200, 150.

Sample Output:



Result :

Thus the distribution of various data science roles is analysed and visualised.

Exp No:2

Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

Aim:

To Upload and Analyze the data set given in csv format and perform data preprocessing and visualization.

Description:

Use sample data set sales-data.csv.

Code:

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
# Load the data into a pandas DataFrame
file_path='C:\sales_data.csv'
df = pd.read_csv(file_path)
# Display the first few rows of the DataFrame
print(df.head())
# Check for missing values
print(df.isnull().sum())
# Fill or drop missing values if necessary
df['Sales'].fillna(df['Sales'].mean(), inplace=True)
df.dropna(subset=['Product', 'Quantity', 'Region'], inplace=True)
# Summary statistics
print(df.describe())
# Group by product and calculate the total sales and quantity
product_summary = df.groupby('Product').agg({
'Sales': 'sum',
```

```
'Quantity': 'sum'
}).reset_index()
print(product_summary)

# Bar plot of total sales by product
plt.figure(figsize=(10, 6))
plt.bar(product_summary['Product'], product_summary['Sales'])
plt.xlabel('Product')
plt.ylabel('Total Sales')
plt.title('Total Sales by Product')
plt.show()
# Line plot of sales over time
df['Date'] = pd.to_datetime(df['Date'])
sales_over_time = df.groupby('Date').agg({'Sales': 'sum'}).reset_index()
plt.figure(figsize=(10, 6))
plt.plot(sales_over_time['Date'],sales_over_time['Sales'])
plt.xlabel('Date')
plt.ylabel('Total Sales')
plt.title('SalesOver Time')
plt.show()
# Pivot table to analyze sales by region and product
pivot_table = df.pivot_table(values='Sales', index='Region', columns='Product',
aggfunc=np.sum, fill_value=0)
print(pivot_table)
# Correlation matrix
correlation_matrix = df.corr()
print(correlation_matrix)
# Heatmap of the correlation matrix
```

```
import seaborn as sns

plt.figure(figsize=(8, 6))

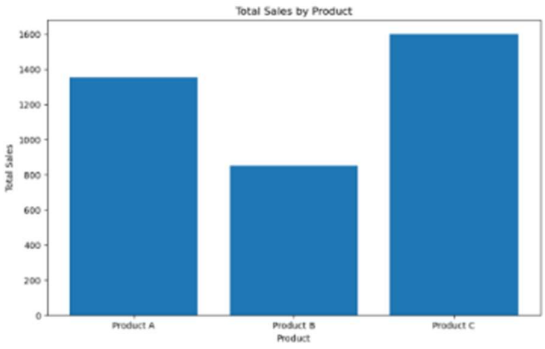
sns.heatmap(correlation_matrix, annot=True, cmap='coolwarm')

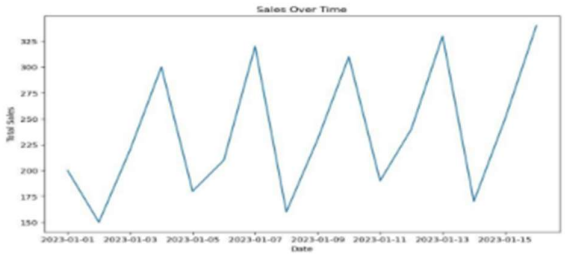
plt.title('Correlation Matrix')

plt.show()
```

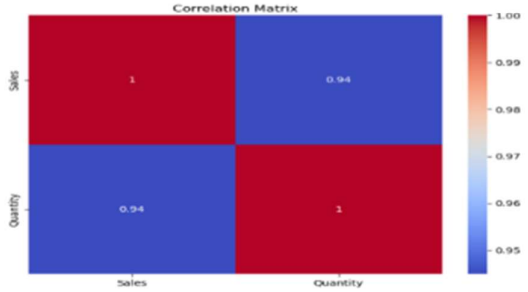
OUTPUT:

```
Date     Product  Sales  Quantity Region
0  2023-01-01  Product A    200         4  North
1  2023-01-02  Product B    150         3  South
2  2023-01-03  Product A    220         5  North
3  2023-01-04  Product C    300         6   East
4  2023-01-05  Product B    180         4   West
Date        0
Product     0
Sales       0
Quantity    0
Region      0
dtype: int64
            Sales   Quantity
count   16.000000  16.000000
mean   237.500000   5.375000
std     64.031242   1.746425
min    150.000000   3.000000
25%    187.500000   4.000000
50%    225.000000   5.500000
75%    302.500000   7.000000
max    340.000000   8.000000
     Product  Sales  Quantity
0  Product A   1350        33
1  Product B    850        17
2  Product C   1600        36
```






```
Product  Product A  Product B  Product C
Region
East             0          0       1600
North         1350          0          0
South            0        480          0
West             0        370          0
               Sales  Quantity
Sales       1.000000  0.944922
Quantity    0.944922  1.000000
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_9648\511106317.py:49: Futur
eWarning: The default value of numeric_only in DataFrame.corr is deprecated
. In a future version, it will default to False. Select only valid columns
or specify the value of numeric_only to silence this warning.
  correlation_matrix = df.corr()
```



Result :

Thus the data set given in csv format and perform data preprocessing and visualization analysed.

Exp No:3.a

Conduct an experiment to show data visualization using line plot

Aim:

Conduct an experiment to show data visualization using line plot

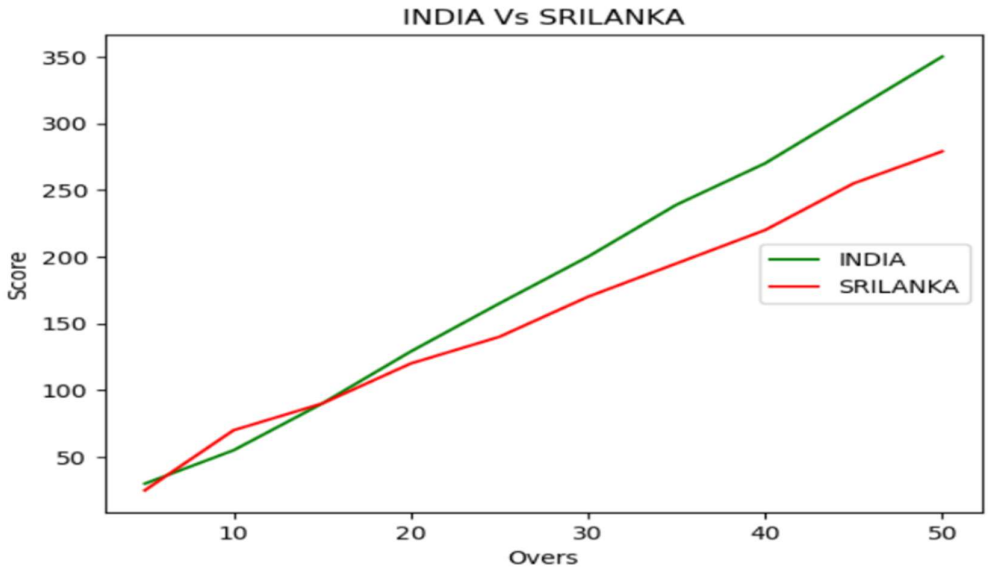
Description:

Take any sample data either through csv file or data fetched directly through code.

Code:

```
import matplotlib.pyplot as cricket
Overs=list(range(5,51,5))
Indian_Score=[30,55,90,129,165,200,239,270,310,350]
Srilankan_Score=[25,70,90,120,140,170,195,220,255,279]
cricket.plot(Overs,Indian_Score)
cricket.plot(Overs,Srilankan_Score)
cricket.show()
cricket.title("INDIA Vs SRILANKA")
cricket.xlabel("Overs")
cricket.ylabel("Score")
cricket.legend()
cricket.plot(Overs,Indian_Score,color="green",label="INDIA")
cricket.plot(Overs,Srilankan_Score,color="red",label="SRILANKA")
cricket.legend(loc="center right")
```

OUTPUT:



Result :

Thus an experiment to show data visualization using line plot is conducted.

Exp No:3.b

Conduct an experiment to show data visualization using bar chart

.

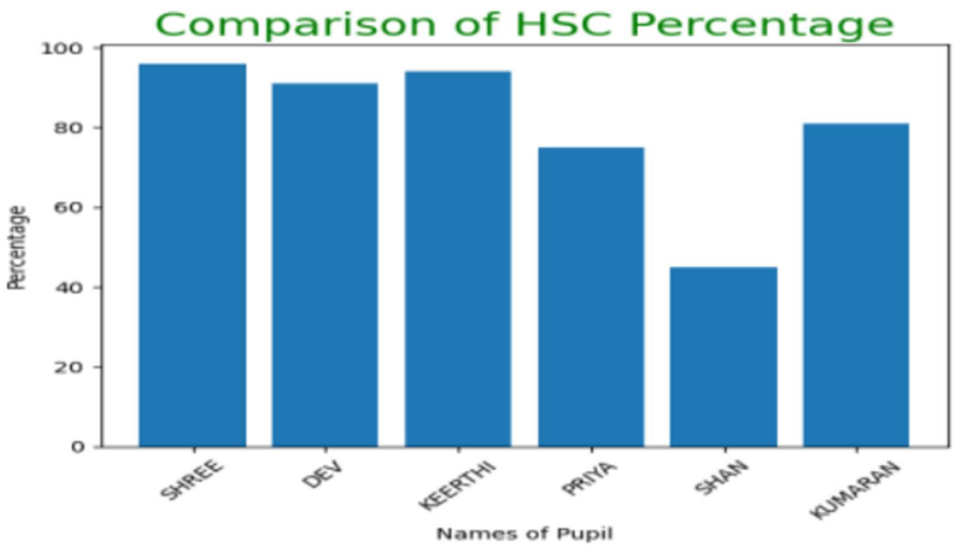
Aim:

Take any sample data either through csv file or data fetched directly through code.

Code :

```
import matplotlib.pyplot as hscmark
import numpy as np
Names = ['SHREE', 'DEV', 'KEERTHI', 'PRIYA', 'SHAN', 'KUMARAN']
xaxis = np.arange(len(Names))
Percentage_hsc = [96, 91, 94, 75, 45, 81]
hscmark.bar(Names, Percentage_hsc)
hscmark.xticks(xaxis, Names, rotation=45)
hscmark.xlabel("Names of Pupil")
hscmark.ylabel("Percentage")
hscmark.title("Comparison of HSC Percentage", fontsize=20, color="green")
hscmark.show()
```

Output :



Result :

Thus an experiment to show data visualization using bar chart is conducted .

Exp No:3.c

Conduct an experiment to show data visualization using pie chart

Aim :

Take any sample data either through csv file or data fetched directly through code.

Code :

```
import matplotlib.pyplot as election

# Election data

labels = ['CANDIDATE 1', 'CANDIDATE 2', 'CANDIDATE 3', 'CANDIDATE 4']

Votes = [315, 130, 245, 210]

colors = ['green', 'yellow', 'red', 'orange']

explode = (0.2, 0, 0, 0)

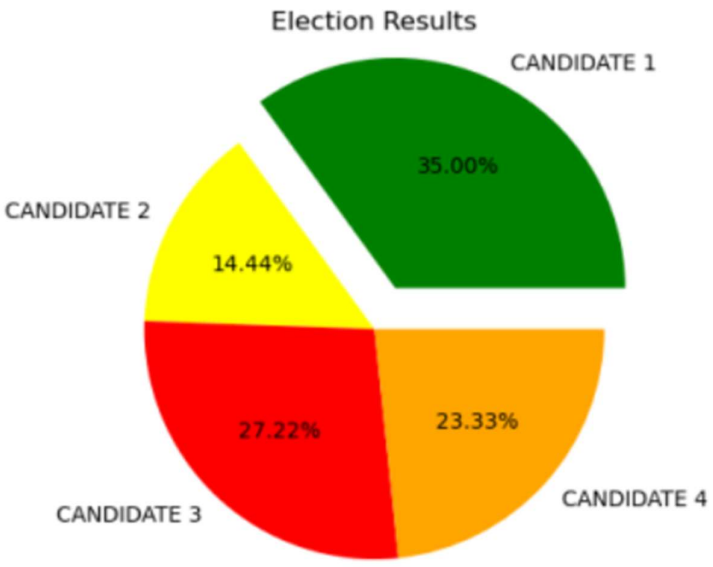
# Plotting the pie chart

election.pie(Votes, labels=labels, colors=colors, explode=explode, autopct='%0.2f%%')

election.title('Election Results')

election.show()
```

Output :



Exp No:4

To Count the frequency of occurrence of a word in a body of text is often needed during text processing.

Aim :

To Import the word_tokenize function and gutenberg.

Code :

```
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import gutenberg
# Download the gutenberg corpus if not already installed
nltk.download('gutenberg')
nltk.download('punkt') # Ensure the punkt tokenizer models are also downloaded
# Load the text from the gutenberg corpus
sample = gutenberg.raw("austen-emma.txt") # Change to an existing text from the corpus
# Tokenize the sample text
token = word_tokenize(sample)
# Create a list of the first 50 tokens
wlist = []
for i in range(50):
wlist.append(token[i])
# Calculate the frequency of each word in the list
wordfreq = [wlist.count(w) for w in wlist]
# Print the word-frequency pairs
print("Pairs\n" + str(list(zip(wlist, wordfreq))))
```

OUTPUT:

Pairs

[('[', 1), ('Emma', 2), ('by', 1), ('Jane', 1), ('Austen', 1), ('1816',

1), (']', 1), ('VOLUME', 1), ('I', 2), ('CHAPTER', 1), ('I', 2), ('Emma

', 2), ('Woodhouse', 1), (',', 5), ('handsome', 1), (',', 5), ('clever'

, 1), (',', 5), ('and', 3), ('rich', 1), (',', 5), ('with', 2), ('a', 1

), ('comfortable', 1), ('home', 1), ('and', 3), ('happy', 1), ('disposi

tion', 1), (',', 5), ('seemed', 1), ('to', 1), ('unite', 1), ('some', 1

), ('of', 2), ('the', 2), ('best', 1), ('blessings', 1), ('of', 2), ('e

xistence', 1), (';', 1), ('and', 3), ('had', 1), ('lived', 1), ('nearly

', 1), ('twenty-one', 1), ('years', 1), ('in', 1), ('the', 2), ('world'

, 1), ('with', 2)]

Result :

Thus the frequency of occurrence of a word in a body of text is counted.

Exp No:5

Data Collection and Initial Exploration

Aim :

To collect, load, and perform initial exploration of the diabetes dataset.

Materials Needed:

- Diabetes dataset (available from sources such as UCI Machine Learning Repository

or Kaggle)

- Python with libraries: pandas, numpy, matplotlib, seaborn

Procedure and Code:

1. Data Collection:

import pandas as pd

Load the dataset

diabetes_df = pd.read_csv(“diabetes.csv”)

Display the first few rows

print(diabetes_df.head())

```
[1] import pandas as pd
    db= pd.read_csv("/content/drive/MyDrive/PIMA Indian Dataset.csv")
```

```
print(db.head())
```

```
   Pregnancies  Glucose  BloodPressure  SkinThickness  Insulin   BMI  \
0            6      148             72             35        0  33.6
1            1       85             66             29        0  26.6
2            8      183             64              0        0  23.3
3            1       89             66             23       94  28.1
4            0      137             40             35      168  43.1

   DiabetesPedigreeFunction  Age  Outcome
0                     0.627   50        1
1                     0.351   31        0
2                     0.672   32        1
3                     0.167   21        0
4                     2.288   33        1
```

2. Initial Exploration:

```python
# Dataset structure and summary statistics
print(db.info())
print(db.describe())

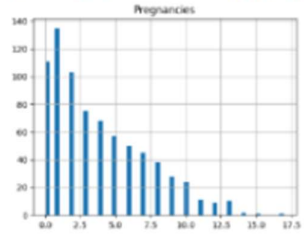
# Visualize the distribution of each feature
import matplotlib.pyplot as plt
import seaborn as sns

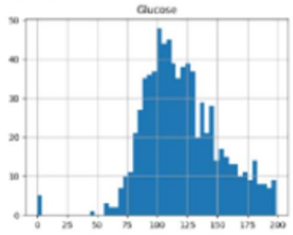
db.hist(bins=50, figsize=(20, 15))
plt.show()

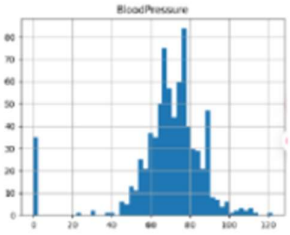
# Pairplot to visualize relationships
sns.pairplot(db)
plt.show()
```

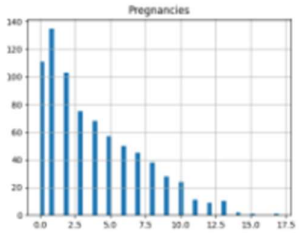
```
[3]
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None
       Pregnancies     Glucose  BloodPressure  SkinThickness     Insulin  \
count   768.000000  768.000000     768.000000     768.000000  768.000000
mean      3.845052  120.894531      69.105469      20.536458   79.799479
std       3.369578   31.972618      19.355807      15.952218  115.244002
min       0.000000    0.000000       0.000000       0.000000    0.000000
25%       1.000000   99.000000      62.000000       0.000000    0.000000
50%       3.000000  117.000000      72.000000      23.000000   30.500000
75%       6.000000  140.250000      80.000000      32.000000  127.250000
max      17.000000  199.000000     122.000000      99.000000  846.000000
```

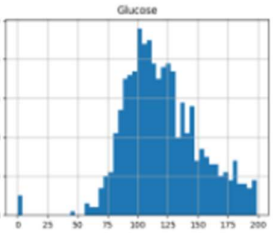
	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.992578	0.471876	33.240885	0.348958
std	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.078000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	32.000000	0.372500	29.000000	0.000000
75%	36.600000	0.626250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000

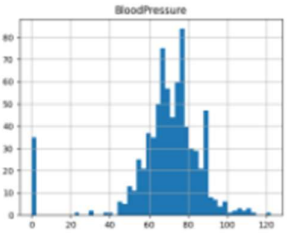


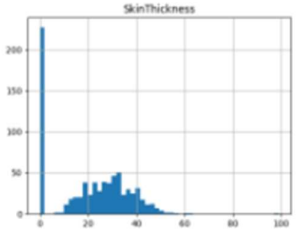


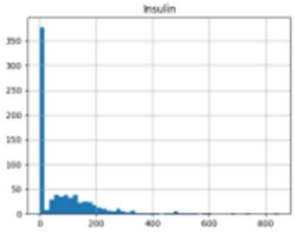


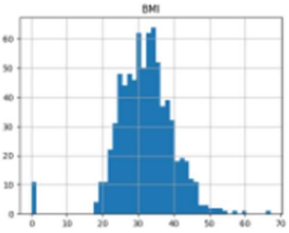


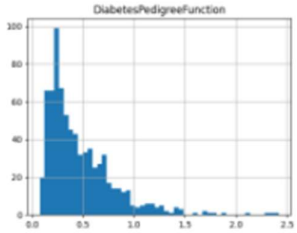


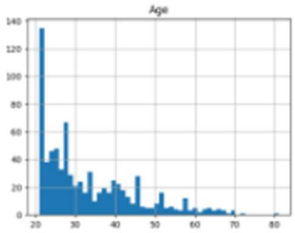


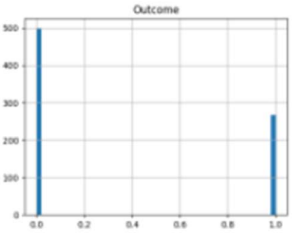


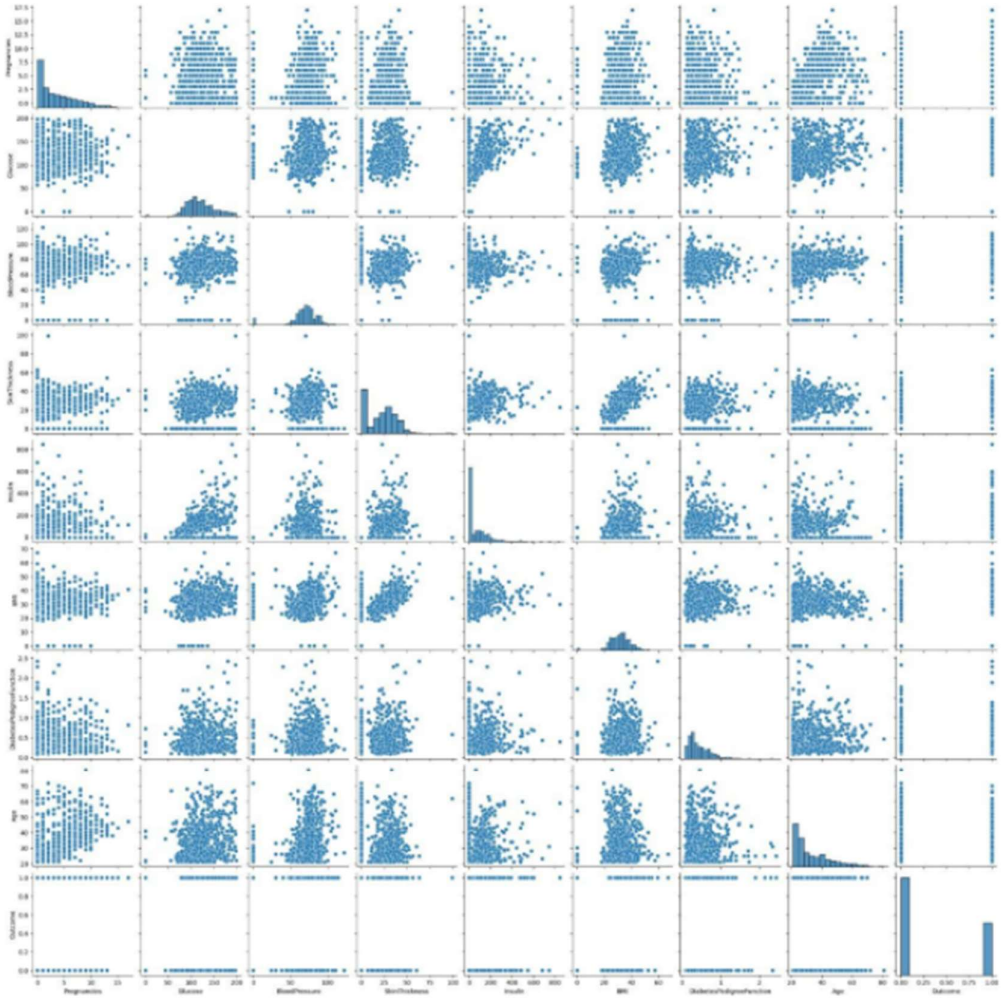












Result :

Thus Data Collection and Initial Exploration is performed.

Exp no : 06

Time Series Analysis of Website Traffic

Aim :

To analyze the time series data of website traffic to identify trends, seasonality, and anomalies.

Dataset:

A dataset containing daily website traffic data with columns such as Date, Page_Views, Unique_Visitors, and Bounce_Rate.

Methodology:

1. Load and Clean Data:

o Import the dataset and parse the Date column as a datetime object.

o Handle missing values by interpolation.

2. Time Series Decomposition:

o Decompose the time series data of Page_Views to identify trend, seasonality, and residuals.

3. Visualization:

o Plot the time series data for Page_Views, Unique_Visitors, and Bounce_Rate.

o Use moving averages to smooth the time series and highlight trends.

o Create seasonal plots to visualize patterns over time.

4. Anomaly Detection:

o Identify and visualize anomalies in the time series data using statistical methods such as the Z-score.

Code :

```
import pandas as pd
import matplotlib.pyplot as plt
```

```
from statsmodels.tsa.seasonal import seasonal_decompose

# Load dataset
data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')

# Time Series Decomposition
decomposition = seasonal_decompose(data['Page_Views'], model='additive')
decomposition.plot()
```

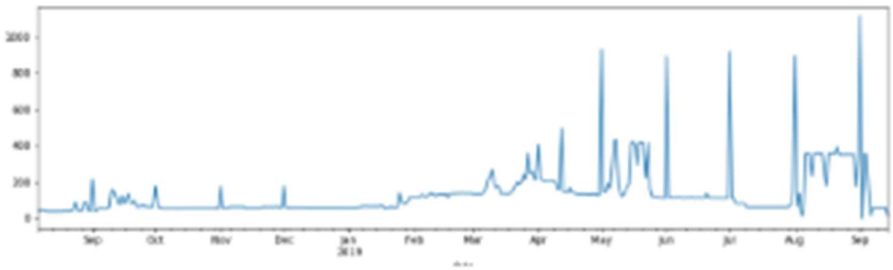


```
plt.show()

# Moving Average
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()
data[['Page_Views', 'Page_Views_MA']].plot()
plt.title('Page Views with Moving Average')
plt.show()

# Anomaly Detection
data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) /
data['Page_Views'].std()
anomalies = data[data['Page_Views_Z'].abs() > 3]
plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')
plt.legend()
plt.show()
```

Output :



Result :

Thus the time series data of website traffic is analysed.

Exp no : 7

Random Sampling and Sampling Distribution

Aim :

To explore random sampling from a population and understand the concept of sampling distribution using Python in Jupyter Notebook.

Steps:

1. Generate a Population:

○ Create a population of data with a specified distribution (e.g., normal distribution).

2. Random Sampling:

○ Perform random sampling from the population to create multiple samples of different sizes.

○ Compute sample statistics (mean, standard deviation, etc.) for each sample.

3. Sampling Distribution:

○ Plot histograms or density plots of sample statistics (e.g., sample means).

○ Compare the sampling distribution of the sample statistic (mean) with the population distribution.

4. Central Limit Theorem (Optional):

○ Demonstrate the Central Limit Theorem by showing that as sample size increases, the sampling distribution of the sample mean approaches a normal distribution regardless of the population distribution.

Code:

```
import numpy as np
import matplotlib.pyplot as plt

# Step 1: Generate a population (e.g., normal distribution)
population_mean = 50
```

```
population_std = 10

population_size = 100000

population = np.random.normal(population_mean, population_std, population_size)


# Step 2: Random sampling

sample_sizes = [30, 50, 100] # different sample sizes to consider

num_samples = 1000 # number of samples for each sample size
```



```
sample_means = {}


for size in sample_sizes:

sample_means[size] = []

for _ in range(num_samples):

sample = np.random.choice(population, size=size, replace=False)

sample_means[size].append(np.mean(sample))


# Step 3: Plotting sampling distributions

plt.figure(figsize=(12, 8))


for i, size in enumerate(sample_sizes):

plt.subplot(len(sample_sizes), 1, i+1)

plt.hist(sample_means[size], bins=30, alpha=0.7, label=f'Sample Size {size}')

plt.axvline(np.mean(population), color='red', linestyle='dashed', linewidth=1.5,
label='Population

Mean')

plt.title(f'Sampling Distribution (Sample Size {size})')
```

```
plt.xlabel('Sample Mean')

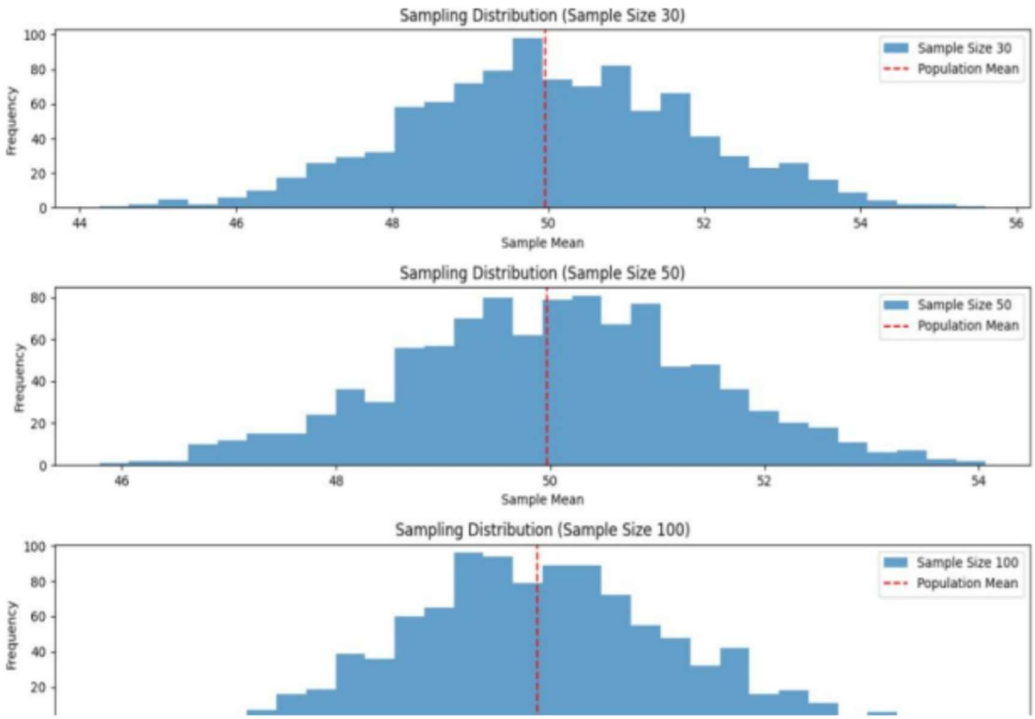
plt.ylabel('Frequency')

plt.legend()


plt.tight_layout()

plt.show()
```

Output :



Result :

Thus Random Sampling and Sampling Distribution concept is understood.

Exp no : 8

Conduct Z test for the Data Given using Python Objective

Aim :

To test whether the average weight of a species of birds differs from 150 grams.

Procedure:

1. Null Hypothesis (H0): The average weight of the birds is 150 grams.

2. Alternative Hypothesis (H1): The average weight of the birds is not 150 grams.

3. Sample: Measure the weights of 30 birds randomly selected from the population.

4. Z-Test: Conduct a Z-test to compare the sample mean to 150 grams.

5. Decision Rule: Use a significance level of α = 0.05.

Code :

```
import numpy as np

import scipy.stats as stats


# Define the sample data (hypothetical weights in grams)

sample_data = np.array([152, 148, 151, 149, 147, 153, 150, 148, 152, 149,151, 150, 149, 152, 151,

148, 150, 152, 149, 150,148, 153, 151, 150, 149, 152, 148, 151, 150, 153])

# Population mean under the null hypothesis

population_mean = 150

# Calculate sample statistics

sample_mean = np.mean(sample_data)

sample_std = np.std(sample_data, ddof=1) # Using sample standard deviation


# Number of observations

n = len(sample_data)

# Calculate the Z-statistic

z_statistic = (sample_mean - population_mean) / (sample_std / np.sqrt(n))
```

```
# Calculate the p-value


p_value = 2 * (1 - stats.norm.cdf(np.abs(z_statistic))) # Two-tailed test


# Print results

print(f"Sample Mean: {sample_mean:.2f}")

print(f"Z-Statistic: {z_statistic:.4f}")

print(f"P-Value: {p_value:.4f}")

# Decision based on the significance level

alpha = 0.05

if p_value < alpha:

print("Reject the null hypothesis: The average weight is significantly different from 150 grams.")

else:

print("Fail to reject the null hypothesis: There is no significant difference in average weight from

150 grams.")
```

OUTPUT:

```
Sample Mean: 150.20

Z-Statistic: 0.6406

P-Value: 0.5218

Fail to reject the null hypothesis: There is no significant difference in average weight from 150 grams.
```

Result :

Thus the Z test for the given is conducted.

Exp No:9

Experiment to understand Matplotlib library use cases in Data Science through visualization

Aim:

Use Iris data set to understand the Matplot lib library

Code :

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline


data=pd.read_csv('/content/Iris_Dataset.csv')
data
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	variety
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
...	...	...	...	...	...	...
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Id             150 non-null    int64
 1   SepalLengthCm  150 non-null    float64
 2   SepalWidthCm   150 non-null    float64
 3   PetalLengthCm  150 non-null    float64
 4   PetalWidthCm   150 non-null    float64
 5   variety        150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

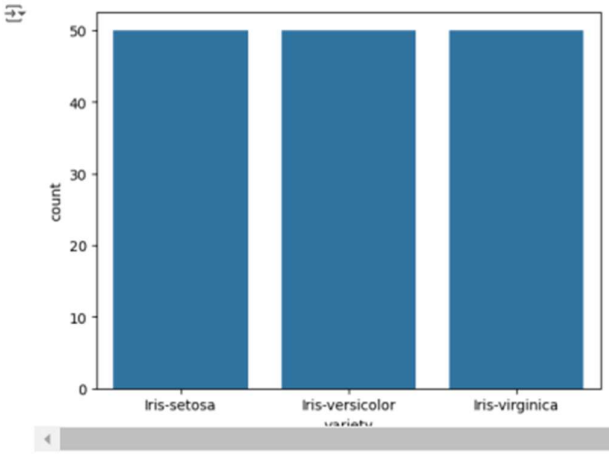
```
data.describe()
```

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
count	150.000000	150.000000	150.000000	150.000000	150.000000
mean	75.500000	5.843333	3.054000	3.758667	1.198667
std	43.445368	0.828066	0.433594	1.764420	0.763161
min	1.000000	4.300000	2.000000	1.000000	0.100000
25%	38.250000	5.100000	2.800000	1.600000	0.300000
50%	75.500000	5.800000	3.000000	4.350000	1.300000
75%	112.750000	6.400000	3.300000	5.100000	1.800000
max	150.000000	7.900000	4.400000	6.900000	2.500000

```
data.value_counts('variety')
```

variety	count
Iris-setosa	50
Iris-versicolor	50
Iris-virginica	50

```
sns.countplot(x='variety',data=data,)
plt.show()
```




```
dummies=pd.get_dummies(data.variety)
```

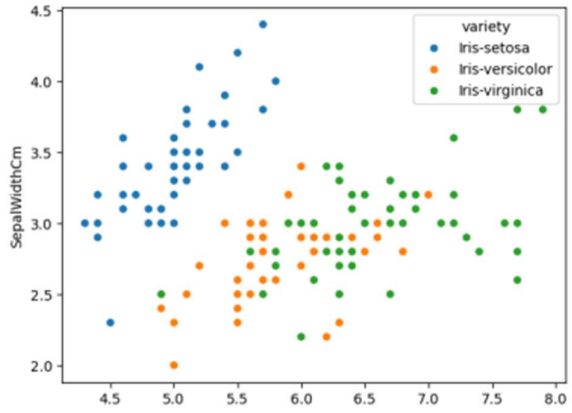
```
FinalDataset=pd.concat([pd.get_dummies(data.variety),data.iloc[:,[0,1,2,3]]],axis=1)
```

```
FinalDataset.head()
```

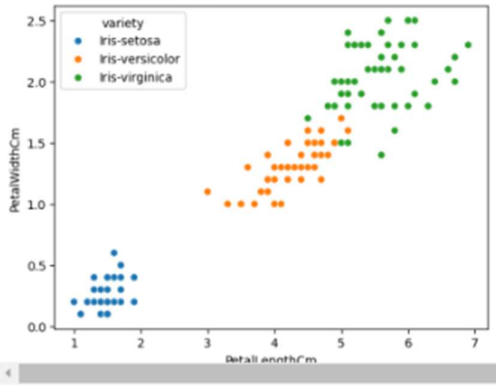
	Iris-setosa	Iris-versicolor	Iris-virginica	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm
0	True	False	False	1	5.1	3.5	1.4
1	True	False	False	2	4.9	3.0	1.4
2	True	False	False	3	4.7	3.2	1.3
3	True	False	False	4	4.6	3.1	1.5
4	True	False	False	5	5.0	3.6	1.4

```
sns.scatterplot(x='SepalLengthCm',y='SepalWidthCm',hue='variety',data=data,)
```

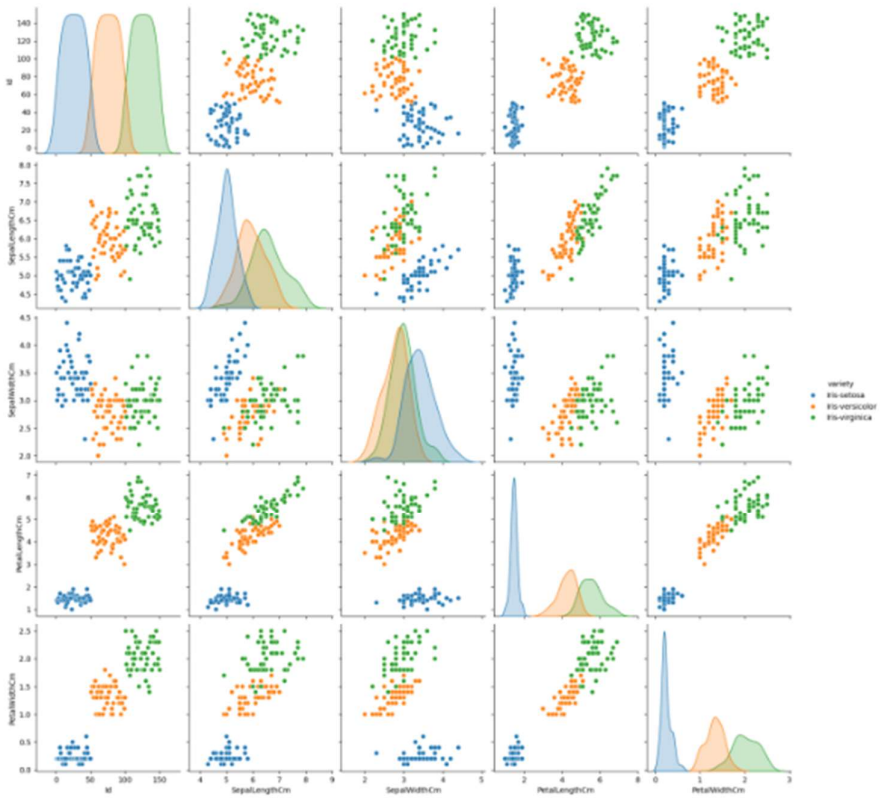
```
<Axes: xlabel='SepalLengthCm', ylabel='SepalWidthCm'>
```



```
<Axes: xlabel='PetalLengthCm', ylabel='PetalWidthCm'>
```

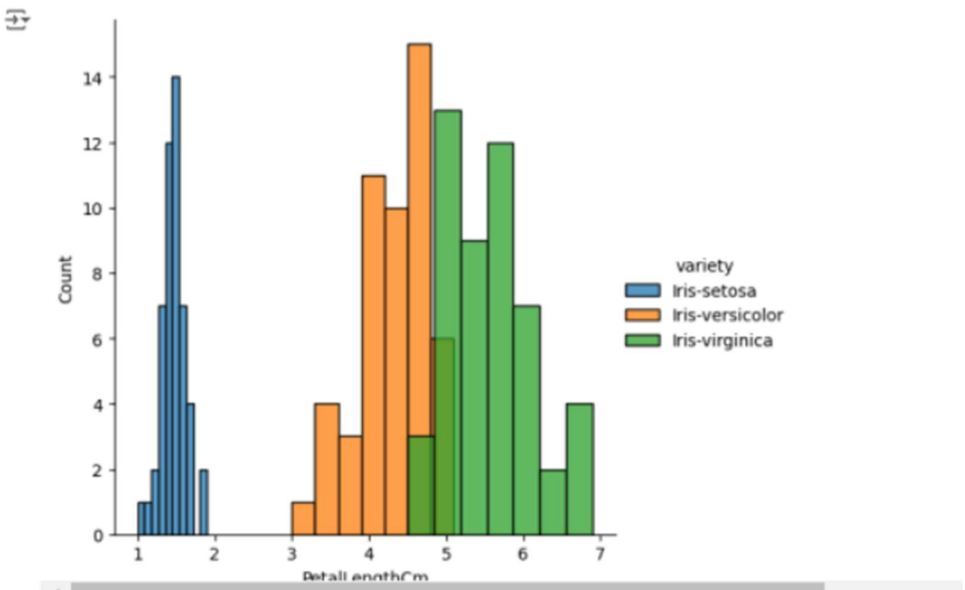



```
sns.pairplot(data,hue='variety',height=3);
```

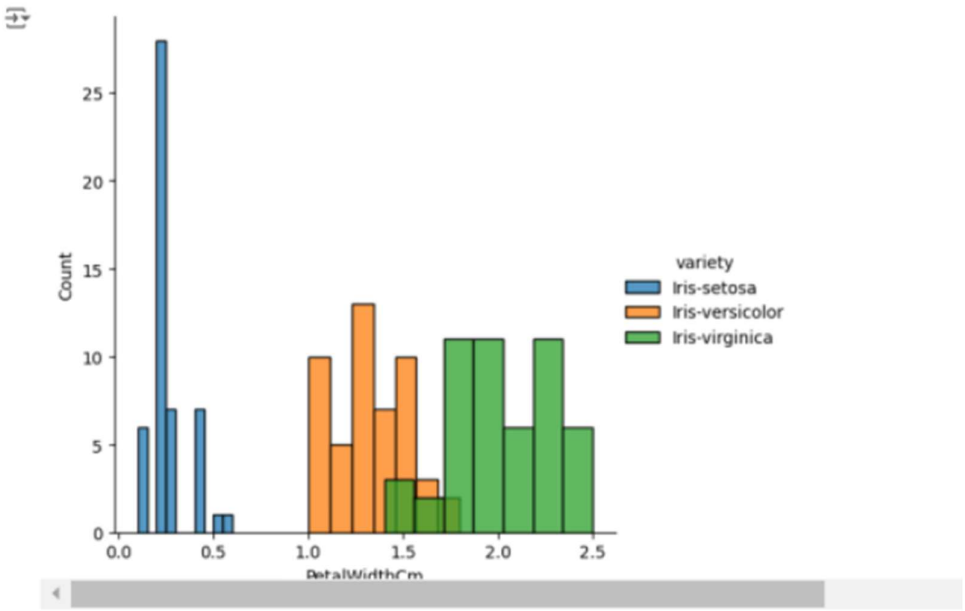


```
plt.show()
```

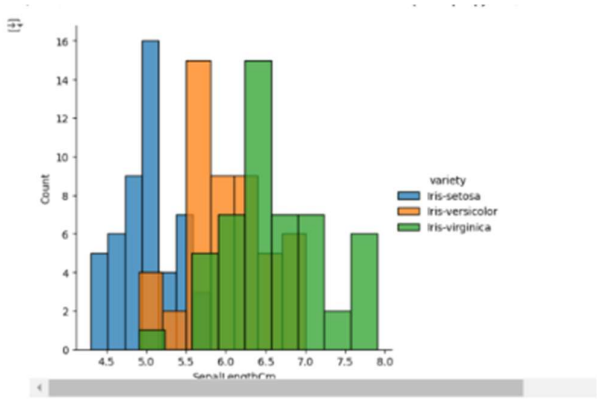
```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalLengthCm').add_legend();
plt.show();
```



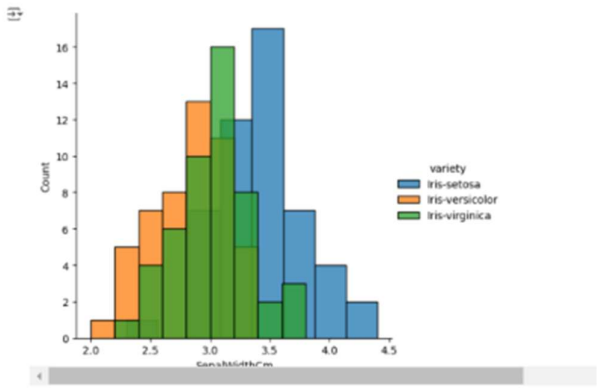

```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'PetalWidthCm').add_legend();
plt.show();
```




```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalLengthCm').add_legend();
plt.show();
```




```
sns.FacetGrid(data,hue='variety',height=5).map(sns.histplot,'SepalWidthCm').add_legend();
plt.show();
```



Result :

Thus the matplotlib library experiment is conducted.

Exp No:10

Experiment to understand array function in Data science

Aim :

Understand array function using Numpy library

Code :

```
import numpy as np
array=np.random.randint(1,100,9)
array
```

```
array([83, 25, 19, 47, 62, 15, 96, 39, 51])
```

```
np.sqrt(array)
```

```
array([9.11043358, 5.        , 4.35889894, 6.8556546 , 7.87400787,
       3.87298335, 9.79795897, 6.244998  , 7.14142843])
```

```
array.ndim
```

```
1
```

```
new_array=array.reshape(3,3)
```

```
new_array
```

```
array([[83, 25, 19],
       [47, 62, 15],
       [96, 39, 51]])
```

```
new_array.ndim
```

```
2
```

```
new_array.ravel()
```

```
array([83, 25, 19, 47, 62, 15, 96, 39, 51])
```

```
newm=new_array.reshape(3,3)
```

```
newm
```

```
array([[83, 25, 19],
       [47, 62, 15],
       [96, 39, 51]])
```

```
newm[2,1:3]
```

```
array([39, 51])
```

```
newm[1:2,1:3]
```

```
array([[62, 15]])
```

```
new_array[0:3,0:0]
```

```
array([], shape=(3, 0), dtype=int64)
```

```
new_array[0:2,0:1]
```

```
array([[83],
       [47]])
```

```
new_array[0:3,0:1]
```

```
array([[83],
       [47],
       [96]])
```

Result :

Thus Experiment to understand array function in Data science is performed.

Exp No:11

Experiment to understand pandas library use cases in Data science.

AIM:

Understand data frame use cases using pandas library

```
import numpy as np
import pandas as pd
list=[[1,'Smith',50000],[2,'Jones',60000]]
```

```
df=pd.DataFrame(list)
df
```

	0	1	2
0	1	Smith	50000
1	2	Jones	60000

```
df.columns=['Empd','Name','Salary']
df
```

	Empd	Name	Salary
0	1	Smith	50000
1	2	Jones	60000

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2 entries, 0 to 1
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Empd    2 non-null      int64
 1   Name    2 non-null      object
 2   Salary  2 non-null      int64
dtypes: int64(2), object(1)
memory usage: 176.0+ bytes
```

```
df=pd.read_csv("/content/50_Startups.csv")
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 50 entries, 0 to 49
Data columns (total 5 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   R&D Spend        50 non-null     float64
 1   Administration   50 non-null     float64
 2   Marketing Spend  50 non-null     float64
 3   State            50 non-null     object 
 4   Profit           50 non-null     float64
dtypes: float64(4), object(1)
memory usage: 2.1+ KB
```

```
df.head()
```

	R&D Spend	Administration	Marketing Spend	State	Profit
0	165349.20	136897.80	471784.10	New York	192261.83
1	162597.70	151377.59	443898.53	California	191792.06
2	153441.51	101145.55	407934.54	Florida	191050.39
3	144372.41	118671.85	383199.62	New York	182901.99
4	142107.34	91391.77	366168.42	Florida	166187.94

```
df.tail()
```

	R&D Spend	Administration	Marketing Spend	State	Profit
45	1000.23	124153.04	1903.93	New York	64926.08
46	1315.46	115816.21	297114.46	Florida	49490.75
47	0.00	135426.92	0.00	California	42559.73
48	542.05	51743.15	0.00	New York	35673.41
49	0.00	116983.80	45173.06	California	14681.40

```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/employee.csv")
```

```
df.head()
```

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000

```
df.tail()
```

	emp id	name	salary
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

```
df.info()
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7 entries, 0 to 6
Data columns (total 3 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   emp id  7 non-null      int64
 1   name    7 non-null      object
 2   salary  7 non-null      int64
dtypes: int64(2), object(1)
memory usage: 296.0+ bytes
```

```
df.salary
```

	salary
0	5000
1	6000
2	7000
3	5000
4	8000
5	3000
6	6000

```
type(df.salary)
```

```
pandas.core.series.Series
def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None,
fastpath: bool=False) -> None

One-dimensional ndarray with axis labels (including time series).

Labels need not be unique but must be a hashable type. The object
supports both integer- and label-based indexing and provides a host of
methods for performing operations involving the index. Statistical
```

```
df.salary.mean()
```

```
5714.285714285715
```

s://colab.research.google.com/drive/1TNEzkVEMxSI_3eUDFZrcEeJH-g7BNg2j

[redacted]

14/24, 12:15 PM panc

```
df.salary.median()
```

```
6000.0
```

```
df.salary.mode()
```

	salary
0	5000
1	6000

```
df.salary.var()
```

```
2571428.5714285714
```

```
df.salary.std()
```

```
1603.5674514745463
```

```
df.describe()
```

	emp id	salary
count	7.000000	7.000000
mean	4.000000	5714.285714
std	2.160247	1603.567451
min	1.000000	3000.000000
25%	2.500000	5000.000000
50%	4.000000	6000.000000
75%	5.500000	6500.000000
max	7.000000	8000.000000

```
df.describe(include='all')
```

	emp id	name	salary
count	7.000000	7	7.000000
unique	NaN	6	NaN
top	NaN	SRI HARSHAVARDHANAN R	NaN
freq	NaN	2	NaN
mean	4.000000	NaN	5714.285714
std	2.160247	NaN	1603.567451
min	1.000000	NaN	3000.000000
25%	2.500000	NaN	5000.000000
50%	4.000000	NaN	6000.000000
75%	5.500000	NaN	6500.000000
max	7.000000	NaN	8000.000000

```
empCol=df.columns
```

```
empCol=df.columns
```

```
empCol
```

```
Index(['emp id', 'name ', 'salary'], dtype='object')
```

```
emparray=df.values
```

```
emparray
```

```
array([[1, 'SREE VARSSINI K S', 5000],
       [2, 'SREEMATHI B', 6000],
       [3, 'SREYA G', 7000],
       [4, 'SREYASKARI MULLAPUDI', 5000],
       [5, 'SRI AKASH U G', 8000],
```

```
          [6, 'SRI HARSHAVARDHANAN R', 3000],
          [7, 'SRI HARSHAVARDHANAN R', 6000]], dtype=object)
```

```
employee_DF=pd.DataFrame(emparray,columns=empCol)
```

```
employee_DF
```

	emp id	name	salary
0	1	SREE VARSSINI K S	5000
1	2	SREEMATHI B	6000
2	3	SREYA G	7000
3	4	SREYASKARI MULLAPUDI	5000
4	5	SRI AKASH U G	8000
5	6	SRI HARSHAVARDHANAN R	3000
6	7	SRI HARSHAVARDHANAN R	6000

```
Start coding or generate with AI.
```

RESULT:

Thus the Experiment to understand pandas library use cases in Data science is done.

Exp No:12

Experiment to detect outliers in a given data set.

AIM:

Understand the procedure to identify the outliers in a given dataset

```
#sample calculation for low range(lr) , upper range (ur),percentile
import numpy as np
array=np.random.randint(1,100,16)  # randomly generate 16 numbers between  1 to 100
array
```

```
array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
array.mean()
```

```
50.5
```

```
np.percentile(array,25)
```

```
26.0
```

```
np.percentile(array,50)
```

```
56.0
```

```
np.percentile(array,75)
```

```
69.0
```

```
np.percentile(array,100)
```

```
90.0
```

```
#outliers detection
def outDetection(array):
  sorted(array)
  Q1,Q3=np.percentile(array,[25,75])
  IQR=Q3-Q1
  lr=Q1-(1.5*IQR)
  ur=Q3+(1.5*IQR)
  return lr,ur

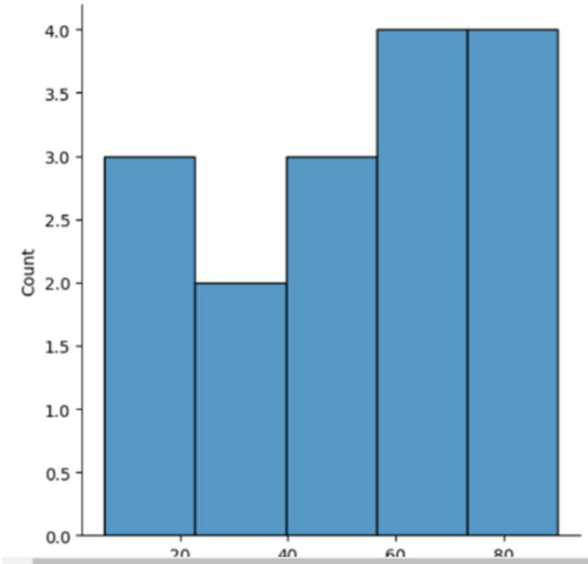
lr,ur=outDetection(array)

lr,ur
```

```
(-38.5, 133.5)
```

```
import seaborn as sns
%matplotlib inline
sns.displot(array)
```

```
<seaborn.axisgrid.FacetGrid at 0x78f3291c2710>
```




```
sns.distplot(array)
```

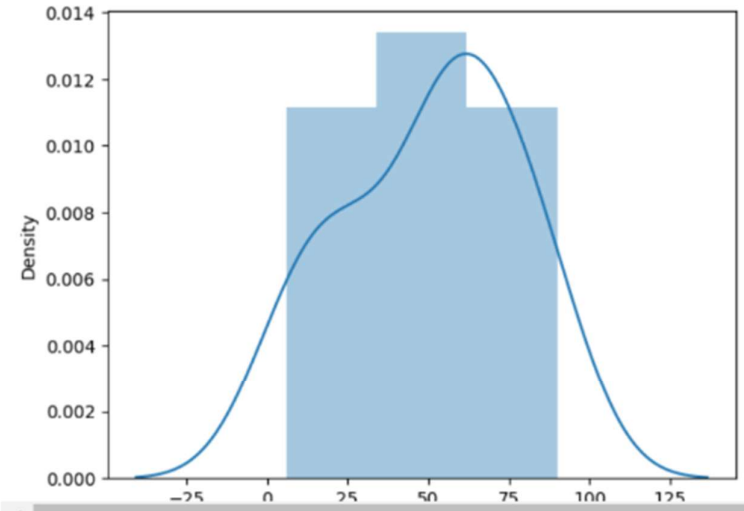
```
<ipython-input-19-d72101983c40>:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(array)
<Axes: ylabel='Density'>
```




```
new_array=array[(array>lr) & (array<ur)]
new_array
```

```
array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

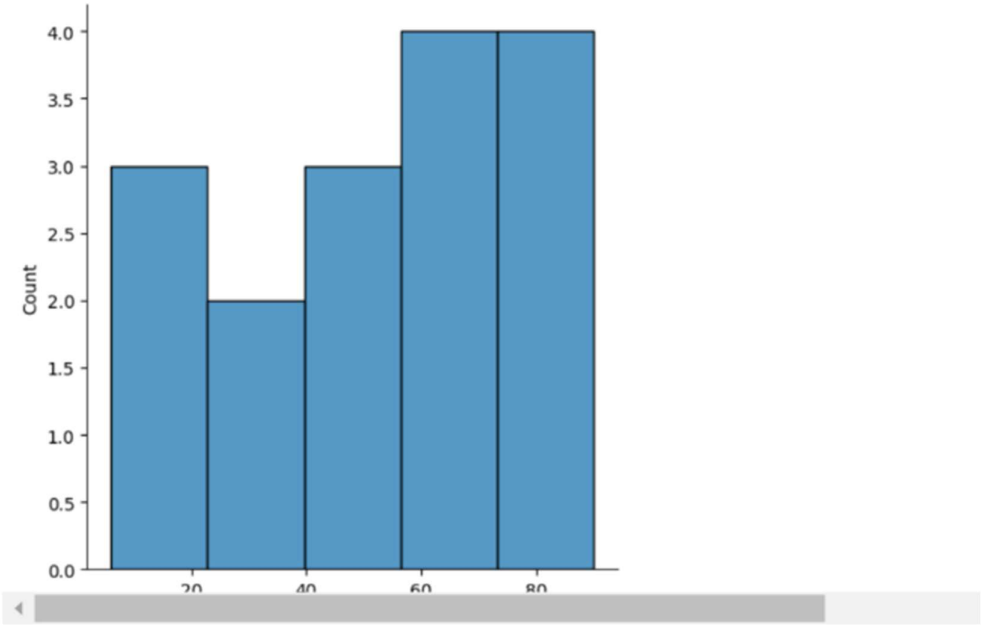
```
sns.displot(new_array)
```

```
new_array=array[(array>lr) & (array<ur)]
new_array
```

```
array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

```
sns.displot(new_array)
```

```
<seaborn.axisgrid.FacetGrid at 0x78f2e09bb580>
```




```
lr1,ur1=outDetection(new_array)
lr1,ur1
```

```
(-38.5, 133.5)
```

```
final_array=new_array[(new_array>lr1) & (new_array<ur1)]
final_array
```

```
array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```

tps://colab.research.google.com/drive/1kQyWP9o5X06QKGZ2THDQgeBxvO2w6OZE#scrollTo=hlPKHYm8_fEK&printMode=true

[redacted]

)/14/24, 1:18 PM Untitled17.ipynb - Colab

```
sns.distplot(final_array)
```

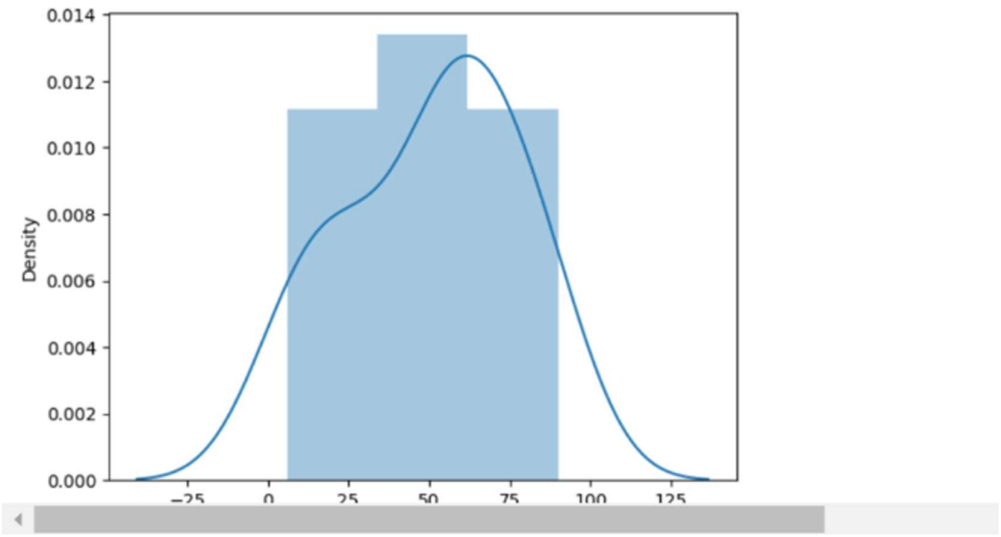
```
<ipython-input-18-7ba96ada5b76>:1: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with
similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

  sns.distplot(final_array)
<Axes: ylabel='Density'>
```



RESULT:

Thus the Experiment to detect outliers in a given data set is determined.

Exp No:13

Experiment to handle missing values and inappropriate data in a given data set

AIM:

Understand the procedure to handle missing and inappropriate data

Dataset Given:

Hotel.csv

CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group
1	20-25	4	Ibis	veg	1300	2	40000	20-25
2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
3	25-30	6	RedFox	Veg	1322	2	30000	25-30
4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
5	35+	3	Ibis	Vegetarian	989	2	45000	35+
6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

About Dataset:

No.of Columns =9 (called as series – CustomerID, Age_Group, Rating(1-5),Hotel, FoodPreference, Bill, NoOfPax, EstimatedSalary)

CutomerID: Numerical Continuous data

Age: Categorical Data

Rating (1-5): Numerical Discrete Data

Hotel: Categorical Data

Food: Categorical Data

Bill: Numerical Continuous data

NoOfPax: Numerical Discrete

EstimatedSalary: Numerical Continuous data

Python Code:

Upload Hotel.csv and convert it into dataFrame

import numpy as np

import pandas as pd

df=pd.read_csv("Hotel_Dataset.csv")

df

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	Estimated Salary	Age_Group.1
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
9	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
10	10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

#From the dataframe identify the duplicate row(i.e row 9)

The duplicated() method returns a Series with True and False values that describe which rows in the DataFrame are duplicated and not.

df.duplicated()

```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9      True
10    False
dtype: bool
```

The info() method prints information about the DataFrame. The information contains the number of columns, column labels, column data types, memory usage, range index, and the number of cells in each column (non-null values).

df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 9 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   CustomerID        11 non-null     int64
 1   Age_Group         11 non-null     object
 2   Rating(1-5)       11 non-null     int64
 3   Hotel             11 non-null     object
 4   FoodPreference    11 non-null     object
 5   Bill              11 non-null     int64
 6   NoOfPax           11 non-null     int64
 7   EstimatedSalary   11 non-null     int64
 8   Age_Group.1       11 non-null     object
dtypes: int64(5), object(4)
memory usage: 924.0+ bytes
```

The drop_duplicates() method removes duplicate rows.

df.drop_duplicates(inplace=True)

df

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group.1
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
10	10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

#While removing duplicate record row index also removed

The len() function to return the length of an object. With a dataframe, the function returns the number of rows.

len(df)

10

#Reset the index

index=np.array(list(range(0,len(df))))

df.set_index(index,inplace=True)

index

```
array([0, 1, 2, 3, 4, 5, 6, 7, 8, 9])
```

df

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary	Age_Group.1
0	1	20-25	4	Ibis	veg	1300	2	40000	20-25
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000	30-35
2	3	25-30	6	RedFox	Veg	1322	2	30000	25-30
3	4	20-25	-1	LemonTree	Veg	1234	2	120000	20-25
4	5	35+	3	Ibis	Vegetarian	989	2	45000	35+
5	6	35+	3	Ibys	Non-Veg	1909	2	122220	35+
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122	35+
7	8	20-25	7	LemonTree	Veg	2999	-10	345673	20-25
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999	25-30
9	10	30-35	5	RedFox	non-Veg	-6755	4	87777	30-35

Axis refers to the dimensions of a DataFrame (index and columns) or Series (index only) Use axis=0 to apply functions row-wise along the index. Use axis=1 to apply functions column-wise across columns.

df.drop(['Age_Group.1'],axis=1,inplace='True)

df

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1	20-25	4	Ibis	veg	1300	2	40000
1	2	30-35	5	LemonTree	Non-Veg	2000	3	59000
2	3	25-30	6	RedFox	Veg	1322	2	30000
3	4	20-25	-1	LemonTree	Veg	1234	2	120000
4	5	35+	3	Ibis	Vegetarian	989	2	45000
5	6	35+	3	Ibys	Non-Veg	1909	2	122220
6	7	35+	4	RedFox	Vegetarian	1000	-1	21122
7	8	20-25	7	LemonTree	Veg	2999	-10	345673
8	9	25-30	2	Ibis	Non-Veg	3456	3	-99999
9	10	30-35	5	RedFox	non-Veg	-6755	4	87777

The function . loc is typically used for label indexing and can access multiple columns.

df.CustomerID.loc[df.CustomerID<0]=np.nan

df.Bill.loc[df.Bill<0]=np.nan

df.EstimatedSalary.loc[df.EstimatedSalary<0]=np.nan

df

```
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2580639570.py:1: S
ettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame
See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df.CustomerID.loc[df.CustomerID<0]=np.nan
df.Bill.loc[df.Bill<0]=np.nan
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2580639570.py:2: S
ettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame
df.EstimatedSalary.loc[df.EstimatedSalary<0]=np.nan
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4.0	Ibis	veg	1300.0	2	40000.0
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3	59000.0
2	3.0	25-30	NaN	RedFox	Veg	1322.0	2	30000.0
3	4.0	20-25	NaN	LemonTree	Veg	1234.0	2	120000.0
4	5.0	35+	3.0	Ibis	Vegetarian	989.0	2	45000.0
5	6.0	35+	3.0	Ibys	Non-Veg	1909.0	2	122220.0
6	7.0	35+	4.0	RedFox	Vegetarian	1000.0	-1	21122.0
7	8.0	20-25	NaN	LemonTree	Veg	2999.0	-10	345673.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3	NaN
9	10.0	30-35	5.0	RedFox	non-Veg	NaN	4	87777.0

df['NoOfPax'].loc[(df['NoOfPax']<1) | (df['NoOfPax']>20)]=np.nan

df

```
C:\Users\Ayyadurai\AppData\Local\Temp\ipykernel_5300\2129877948.py:1: S
ettingWithCopyWarning:
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-
docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy
  df['NoOfPax'].loc[(df['NoOfPax']<1) | (df['NoOfPax']>20)]=np.nan
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4.0	Ibis	veg	1300.0	2.0	40000.0
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3.0	25-30	NaN	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-25	NaN	LemonTree	Veg	1234.0	2.0	120000.0
4	5.0	35+	3.0	Ibis	Vegetarian	989.0	2.0	45000.0
5	6.0	35+	3.0	Ibys	Non-Veg	1909.0	2.0	122220.0
6	7.0	35+	4.0	RedFox	Vegetarian	1000.0	NaN	21122.0
7	8.0	20-25	NaN	LemonTree	Veg	2999.0	NaN	345673.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3.0	NaN
9	10.0	30-35	5.0	RedFox	non-Veg	NaN	4.0	87777.0

df.Age_Group.unique()

```
array(['20-25', '30-35', '25-30', '35+'], dtype=object)
```

df.Hotel.unique()

```
array(['Ibis', 'LemonTree', 'RedFox', 'Ibys'], dtype=object)
```

Using the inplace=True keyword in a pandas method changes the default behaviour such that the operation on the dataframe doesn't return anything, it instead 'modifies the underlying data

df.Hotel.replace(['Ibys'],'Ibis',inplace=True)

df.FoodPreference.unique

```
<bound method Series.unique of 0           veg
1       Non-Veg
2           Veg
3           Veg
4    Vegetarian
5       Non-Veg
6    Vegetarian
7           Veg
8       Non-Veg
9       non-Veg
Name: FoodPreference, dtype: object>
```

df.FoodPreference.replace(['Vegetarian','veg'],'Veg',inplace=True)

df.FoodPreference.replace(['non-Veg'],'Non-Veg',inplace=True)

Fillna is a Pandas function to fill the NA/NaN values with the specified method.

If column or feature is numerical continuous data then replace the missing(NaN) value by taking mean value.

If column or feature is numerical discrete data then replace the missing(NaN) value by taking median value.

If column or feature is non-numerical i.e Categorical data then replace the missing(NaN) value by taking mode value.

```
df.EstimatedSalary.fillna(round(df.EstimatedSalary.mean()),inplace=True)

df.NoOfPax.fillna(round(df.NoOfPax.median()),inplace=True)

df['Rating(1-5)'].fillna(round(df['Rating(1-5)'].median()), inplace=True)

df.Bill.fillna(round(df.Bill.mean()),inplace=True)

df
```

	CustomerID	Age_Group	Rating(1-5)	Hotel	FoodPreference	Bill	NoOfPax	EstimatedSalary
0	1.0	20-25	4.0	Ibis	Veg	1300.0	2.0	40000.0
1	2.0	30-35	5.0	LemonTree	Non-Veg	2000.0	3.0	59000.0
2	3.0	25-30	4.0	RedFox	Veg	1322.0	2.0	30000.0
3	4.0	20-25	4.0	LemonTree	Veg	1234.0	2.0	120000.0
4	5.0	35+	3.0	Ibis	Veg	989.0	2.0	45000.0
5	6.0	35+	3.0	Ibis	Non-Veg	1909.0	2.0	122220.0
6	7.0	35+	4.0	RedFox	Veg	1000.0	2.0	21122.0
7	8.0	20-25	4.0	LemonTree	Veg	2999.0	2.0	345673.0
8	9.0	25-30	2.0	Ibis	Non-Veg	3456.0	3.0	96755.0
9	10.0	30-35	5.0	RedFox	Non-Veg	1801.0	4.0	87777.0

Result :

Thus Experiment to handle missing values and inappropriate data in a given data set is performed.

Exp No:14

Experiment to understand feature scaling.

Aim:

Understand the importance of feature scaling

Code:

```
import numpy as np
import pandas as pd
df=pd.read_csv('/content/pre-process_datasample.csv')
```

```
df
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	NaN	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df.head()
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes

Next steps: Generate code with df | View recommended plots | New interactive sheet

```
df.Country.fillna(df.Country.mode()[0],inplace=True)
features=df.iloc[:,:-1].values
```

```
<ipython-input-5-20665a0bbaa1>:1: FutureWarning: A value is trying to be set on a copy of a DataFrame c
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate ob

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inpla


  df.Country.fillna(df.Country.mode()[0],inplace=True)
```

```
age.fit(features[:,[1]])
```

```
SimpleImputer()
```

```
Salary.fit(features[:,[2]])
```

```
SimpleImputer()
```

```
SimpleImputer()
```

```
features[:,[1]]=age.transform(features[:,[1]])


features[:,[2]]=Salary.transform(features[:,[2]])


features
```

```
array([['France', 44.0, 72000.0],
       ['Spain', 27.0, 48000.0],
       ['Germany', 30.0, 54000.0],
       ['Spain', 38.0, 61000.0],
       ['Germany', 40.0, 63777.77777777778],
       ['France', 35.0, 58000.0],
       ['Spain', 38.77777777777778, 52000.0],
       ['France', 48.0, 79000.0],
       ['France', 50.0, 83000.0],
       ['France', 37.0, 67000.0]], dtype=object)
```

```
from sklearn.preprocessing import OneHotEncoder


oh = OneHotEncoder(sparse_output=False)


Country=oh.fit_transform(features[:,[0]])


Country
```

```
array([[1., 0., 0.],
       [0., 0., 1.],
       [0., 1., 0.],
       [0., 0., 1.],
       [0., 1., 0.],
       [1., 0., 0.],
       [0., 0., 1.],
       [1., 0., 0.],
```

```
final_set
```

```
array([[1.0, 0.0, 0.0, 44.0, 72000.0],
       [0.0, 0.0, 1.0, 27.0, 48000.0],
       [0.0, 1.0, 0.0, 30.0, 54000.0],
       [0.0, 0.0, 1.0, 38.0, 61000.0],
       [0.0, 1.0, 0.0, 40.0, 63777.77777777778],
       [1.0, 0.0, 0.0, 35.0, 58000.0],
       [0.0, 0.0, 1.0, 38.77777777777778, 52000.0],
       [1.0, 0.0, 0.0, 48.0, 79000.0],
       [1.0, 0.0, 0.0, 50.0, 83000.0],
       [1.0, 0.0, 0.0, 37.0, 67000.0]], dtype=object)
```

```
from sklearn.preprocessing import StandardScaler
sc=StandardScaler()
sc.fit(final_set)
feat_standard_scaler=sc.transform(final_set)
```

```
feat_standard_scaler
```

```
array([[ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
         7.58874362e-01,  7.49473254e-01],
       [-1.00000000e+00, -5.00000000e-01,  1.52752523e+00,
        -1.71150388e+00, -1.43817841e+00],
       [-1.00000000e+00,  2.00000000e+00, -6.54653671e-01,
        -1.27555478e+00, -8.91265492e-01],
       [-1.00000000e+00, -5.00000000e-01,  1.52752523e+00,
        -1.13023841e-01, -2.53200424e-01],
       [-1.00000000e+00,  2.00000000e+00, -6.54653671e-01,
         1.77608893e-01,  6.63219199e-16],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
        -5.48972942e-01, -5.26656882e-01],
       [-1.00000000e+00, -5.00000000e-01,  1.52752523e+00,
         0.00000000e+00, -1.07356980e+00],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
         1.34013983e+00,  1.38753832e+00],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
         1.63077256e+00,  1.75214693e+00],
       [ 1.00000000e+00, -5.00000000e-01, -6.54653671e-01,
        -2.58340208e-01,  2.93712492e-01]])
```

```
from sklearn.preprocessing import MinMaxScaler
mms=MinMaxScaler(feature_range=(0,1))
mms.fit(final_set)
feat_minmax_scaler=mms.transform(final_set)
feat_minmax_scaler
```

```
array([[1.        , 0.        , 0.        , 0.73913043, 0.68571429],
       [0.        , 0.        , 1.        , 0.        , 0.        ],
       [0.        , 1.        , 0.        , 0.13043478, 0.17142857],
       [0.        , 0.        , 1.        , 0.47826087, 0.37142857],
       [0.        , 1.        , 0.        , 0.56521739, 0.45079365],
       [1.        , 0.        , 0.        , 0.34782609, 0.28571429],
       [0.        , 0.        , 1.        , 0.51207729, 0.11428571],
       [1.        , 0.        , 0.        , 0.91304348, 0.88571429],
       [1.        , 0.        , 0.        , 1.        , 1.        ],
       [1.        , 0.        , 0.        , 0.43478261, 0.54285714]])
```

Result :

Thus the feature scaling experiment is performed.

Exp No:15

Experiment to understand the data preprocessing in Data science

AIM:

Understand the importance of Data preprocessing in data science

CODE:

```
import numpy as np
import pandas as pd
df=pd.read_csv("/content/pre-process_datasample.csv")
df
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	NaN	Yes
5	France	35.0	58000.0	Yes
6	Spain	NaN	52000.0	No
7	France	48.0	79000.0	Yes
8	NaN	50.0	83000.0	No
9	France	37.0	67000.0	Yes

Double-click (or enter) to edit

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype
---  ------     --------------  -----
 0   Country    9 non-null      object
 1   Age        9 non-null      float64
 2   Salary     9 non-null      float64
 3   Purchased  10 non-null     object
dtypes: float64(2), object(2)
memory usage: 448.0+ bytes
```

```
df.Country.mode()
```

	Country
0	France

```
df.Country.mode()[0]
```

```
type(df.Country.mode())
```

```
pandas.core.series.Series
def __init__(data=None, index=None, dtype: Dtype | None=None, name=None, copy: bool | None=None,
fastpath: bool=False) -> None
    index is not None, the resulting Series is reindexed with the index values.
dtype : str, numpy.dtype, or ExtensionDtype, optional
    Data type for the output Series. If not specified, this will be
    inferred from `data`.
    See the :ref:`user guide <basics.dtypes>` for more usages.
name : Hashable, default None
    The name to give to the Series.
```

```
df.Country.fillna(df.Country.mode()[0],inplace=True)
```

```
df.Age.fillna(df.Age.median(),inplace=True)
```

```
df.Salary.fillna(round(df.Salary.mean()),inplace=True)
```

```
df
```

	Country	Age	Salary	Purchased
0	France	44.0	72000.0	No
1	Spain	27.0	48000.0	Yes
2	Germany	30.0	54000.0	No
3	Spain	38.0	61000.0	No
4	Germany	40.0	63778.0	Yes
5	France	35.0	58000.0	Yes
6	Spain	38.0	52000.0	No
7	France	48.0	79000.0	Yes
8	France	50.0	83000.0	No
9	France	37.0	67000.0	Yes

```
pd.get_dummies(df.Country)
```

	France	Germany	Spain
0	True	False	False
1	False	False	True
2	False	True	False
3	False	False	True
4	False	True	False
5	True	False	False
6	False	False	True
7	True	False	False
8	True	False	False
9	True	False	False

```
updated_dataset=pd.concat([pd.get_dummies(df.Country),df.iloc[:,[1,2,3]]],axis=1)
```

```
updated_dataset
```

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	No
1	False	False	True	27.0	48000.0	Yes
2	False	True	False	30.0	54000.0	No
3	False	False	True	38.0	61000.0	No
4	False	True	False	40.0	63778.0	Yes
5	True	False	False	35.0	58000.0	Yes
6	False	False	True	38.0	52000.0	No
7	True	False	False	48.0	79000.0	Yes
8	True	False	False	50.0	83000.0	No
9	True	False	False	37.0	67000.0	Yes

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10 entries, 0 to 9
Data columns (total 4 columns):
 #   Column     Non-Null Count  Dtype  
---  ------     --------------  -----  
 0   Country    10 non-null     object 
 1   Age        10 non-null     float64
 2   Salary     10 non-null     float64
 3   Purchased  10 non-null     object 
dtypes: float64(2), object(2)
memory usage: 448.0+ bytes
```

```
updated_dataset.Purchased.replace(['No','Yes'],[0,1],inplace=True)
```

```
updated_dataset
```

	France	Germany	Spain	Age	Salary	Purchased
0	True	False	False	44.0	72000.0	0
1	False	False	True	27.0	48000.0	1
2	False	True	False	30.0	54000.0	0
3	False	False	True	38.0	61000.0	0
4	False	True	False	40.0	63778.0	1
5	True	False	False	35.0	58000.0	1
6	False	False	True	38.0	52000.0	0
7	True	False	False	48.0	79000.0	1
8	True	False	False	50.0	83000.0	0
9	True	False	False	37.0	67000.0	1

RESULT :

Thus the data preprocessing experiment in data science is done.

Exp No:16

Experiment to understand EDA-Quantitative and Qualitative analysis.

AIM:

Understand the importance of EDA-Quantitative and Qualitative analysis.

CODE:

```
import seaborn as sns
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
%matplotlib inline

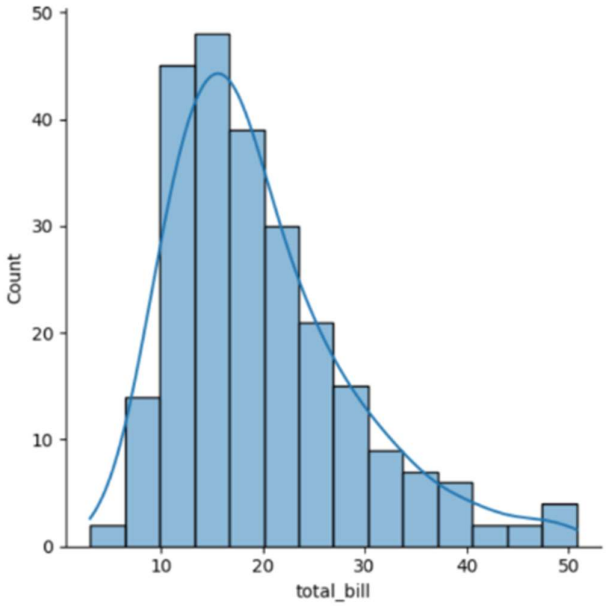

tips=sns.load_dataset('tips')


tips.head()
```

	total_bill	tip	sex	smoker	day	time	size
0	16.99	1.01	Female	No	Sun	Dinner	2
1	10.34	1.66	Male	No	Sun	Dinner	3
2	21.01	3.50	Male	No	Sun	Dinner	3
3	23.68	3.31	Male	No	Sun	Dinner	2
4	24.59	3.61	Female	No	Sun	Dinner	4

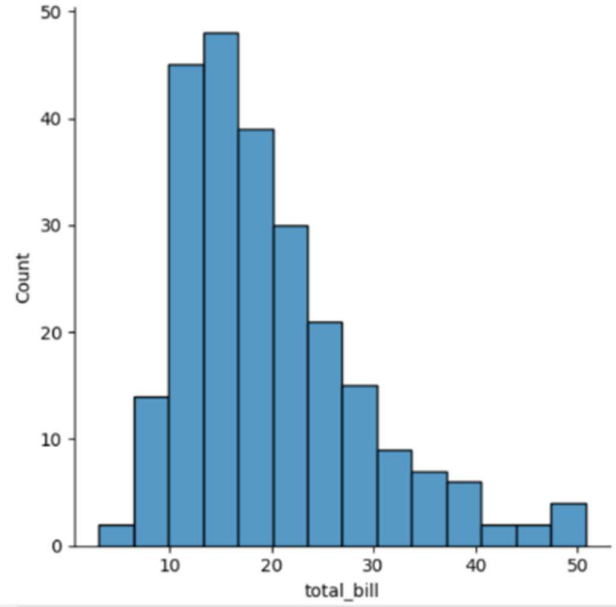
```
sns.displot(tips.total_bill,kde=True)
```

```
<seaborn.axisgrid.FacetGrid at 0x79bb4c7ea680>
```



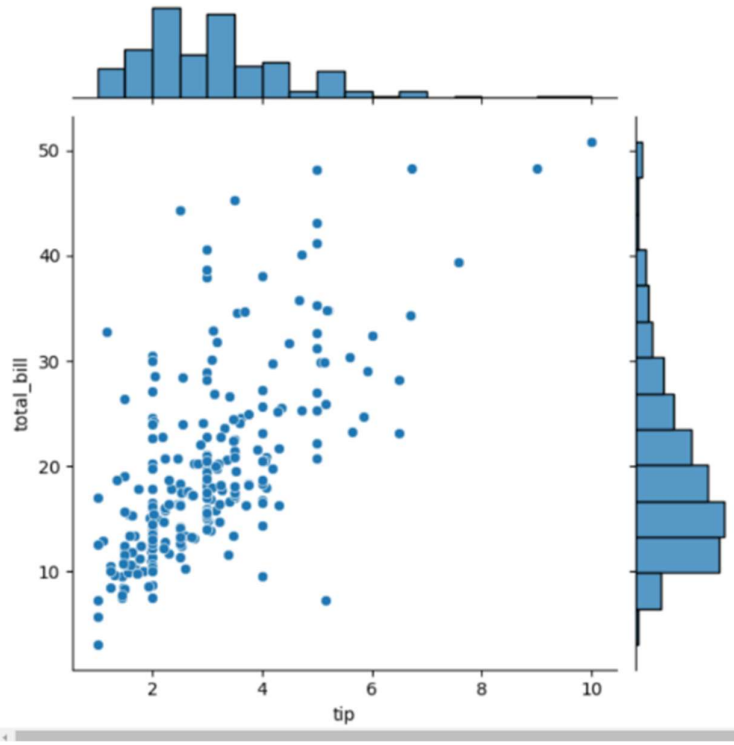

```
sns.displot(tips.total_bill,kde=False)
```

```
<seaborn.axisgrid.FacetGrid at 0x79bb0b0af580>
```

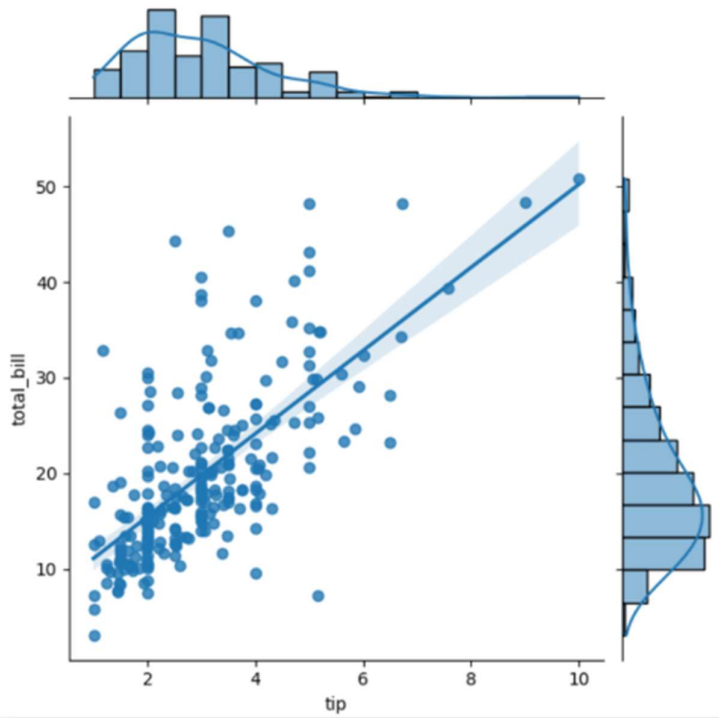



```
sns.jointplot(x=tips.tip,y=tips.total_bill)
```

```
<seaborn.axisgrid.JointGrid at 0x79bb08fc96c0>
```

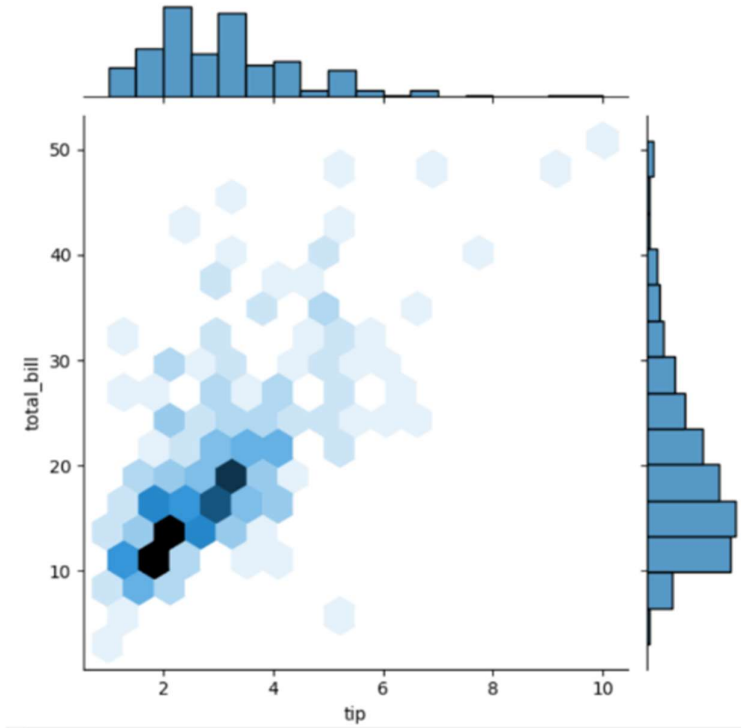


<seaborn.axisgrid.JointGrid at 0x79bb08fc9cf0>



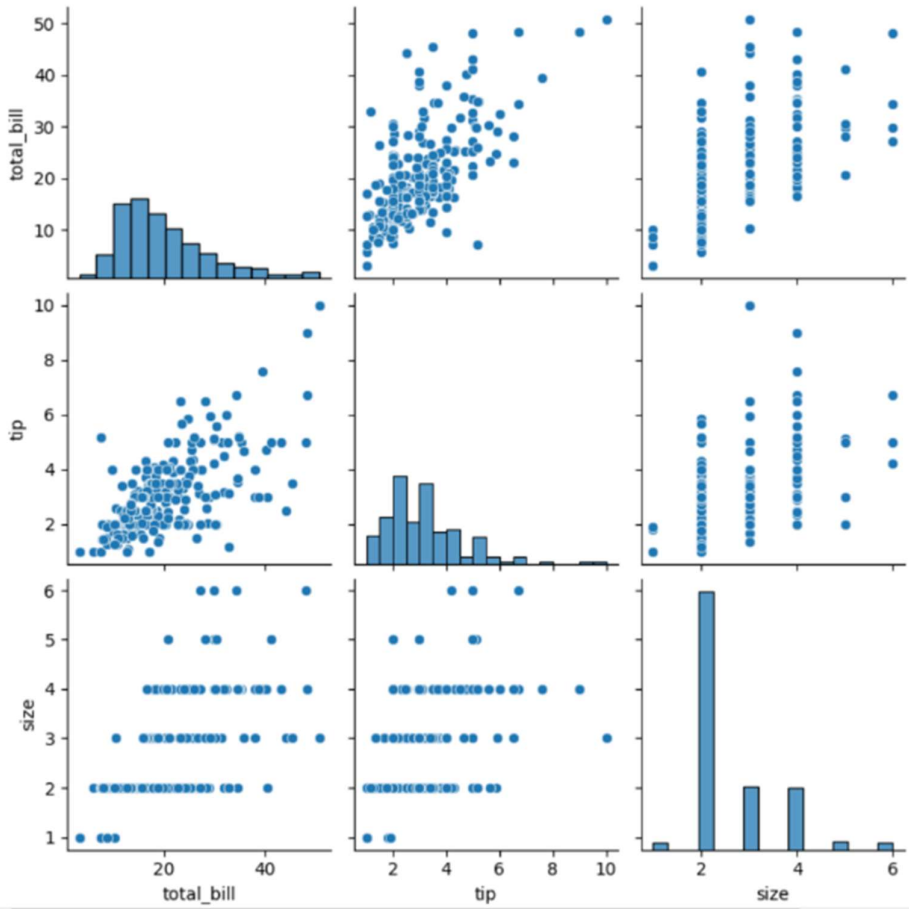

```
sns.jointplot(x=tips.tip,y=tips.total_bill,kind="hex")
```

<seaborn.axisgrid.JointGrid at 0x79bb088f4730>

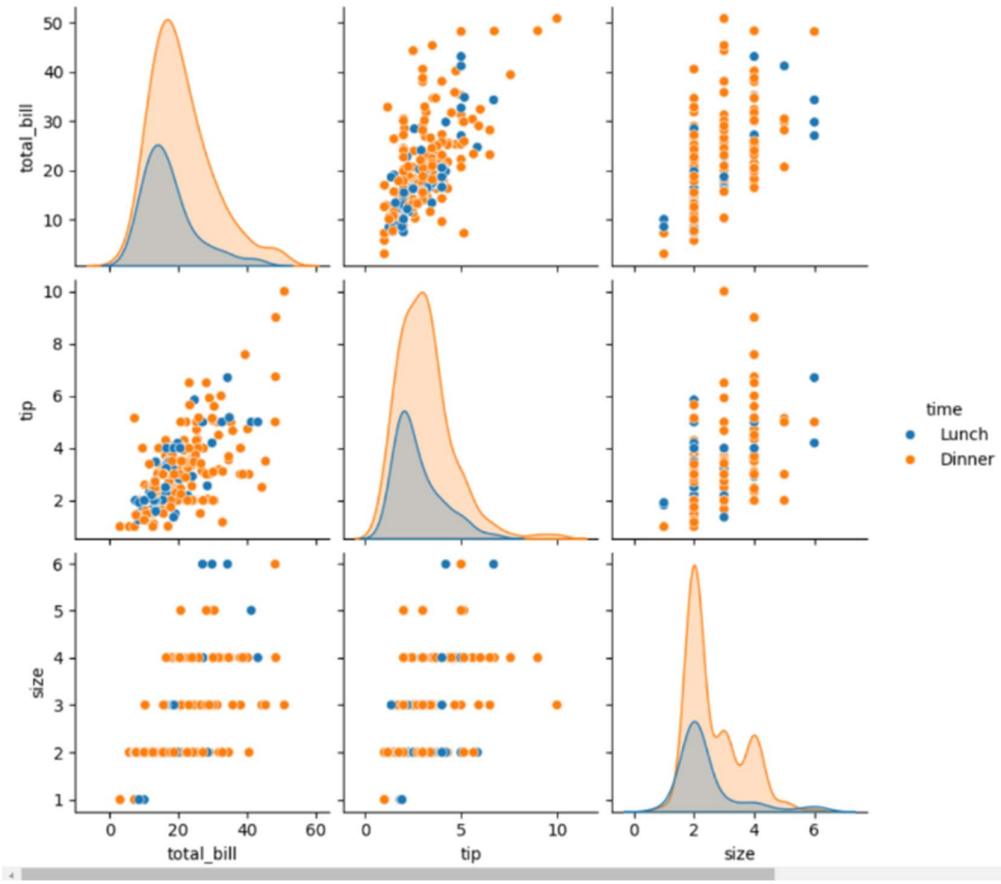



```
sns.pairplot(tips)
```

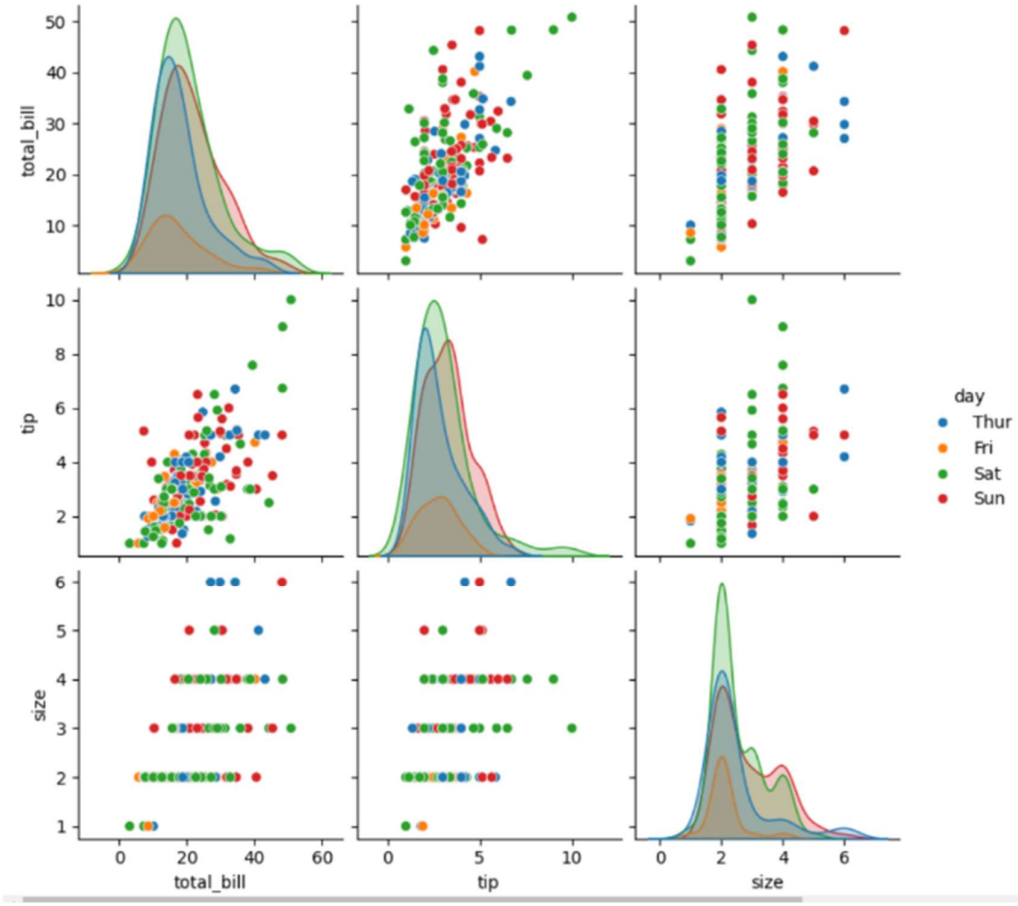
```
<seaborn.axisgrid.PairGrid at 0x79bb06fc3d30>
```



```
<seaborn.axisgrid.PairGrid at 0x79bb088f4670>
```

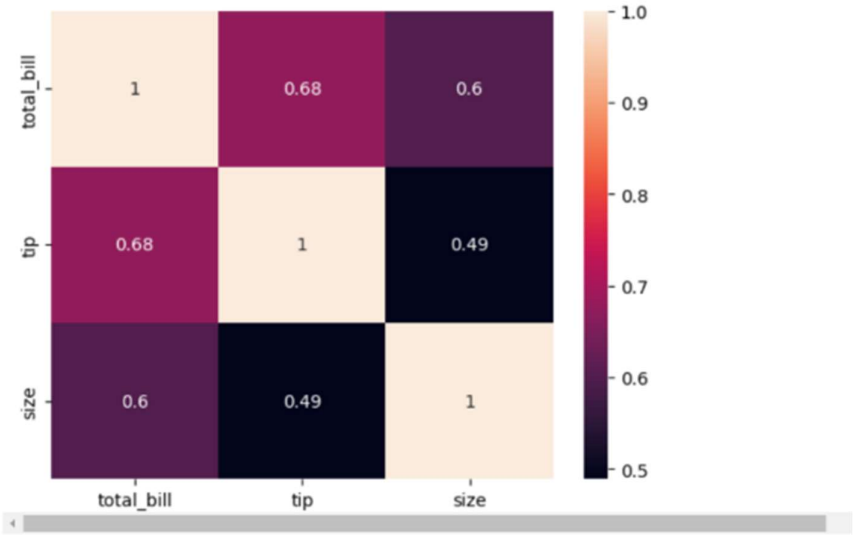


```
<seaborn.axisgrid.PairGrid at 0x79bb08f1f6a0>
```



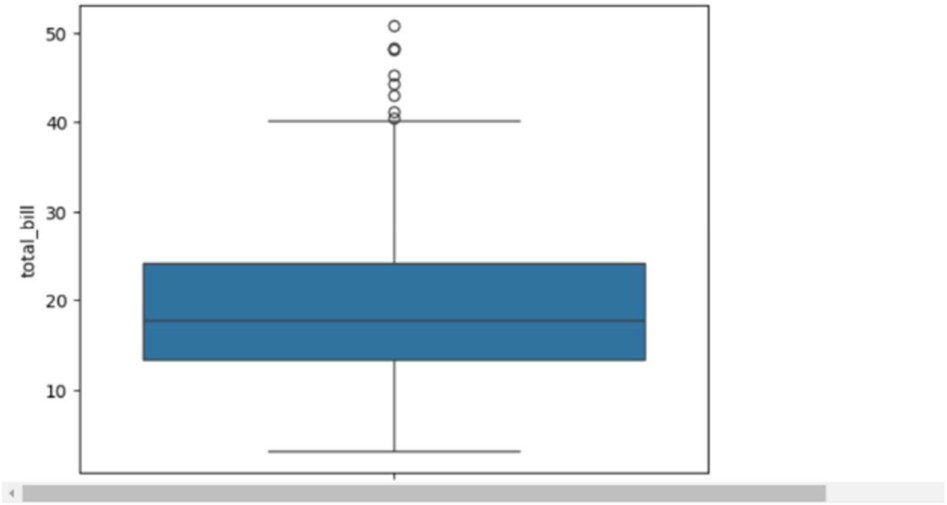

```
sns.heatmap(tips.corr(numeric_only=True),annot=True)
```

```
<Axes: >
```



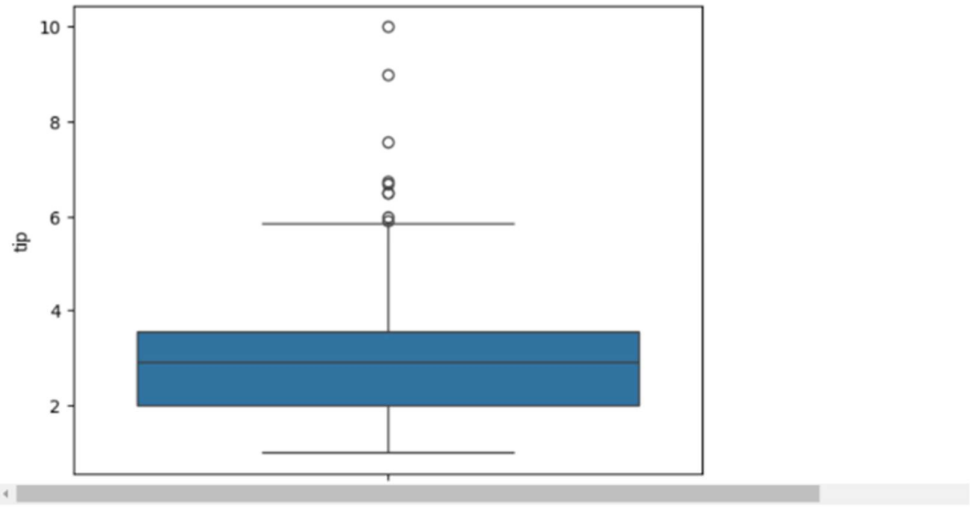

```
sns.boxplot(tips.total_bill)
```

<Axes: ylabel='total_bill'>




```
sns.boxplot(tips.tip)
```

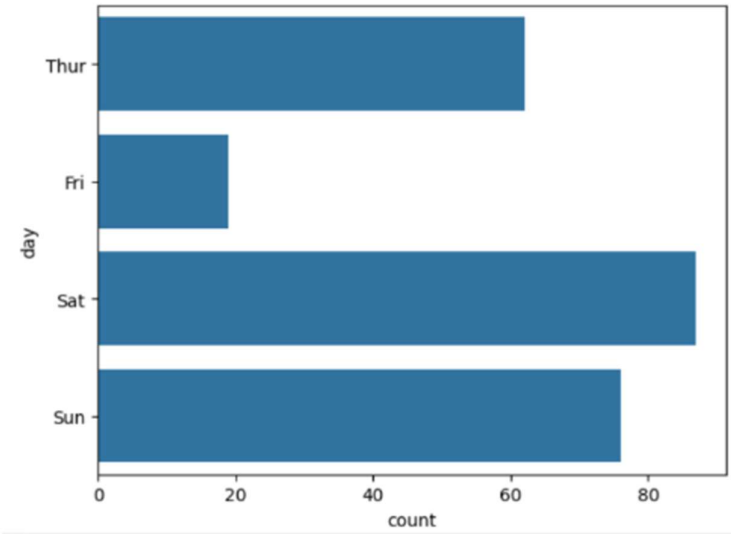
<Axes: ylabel='tip'>



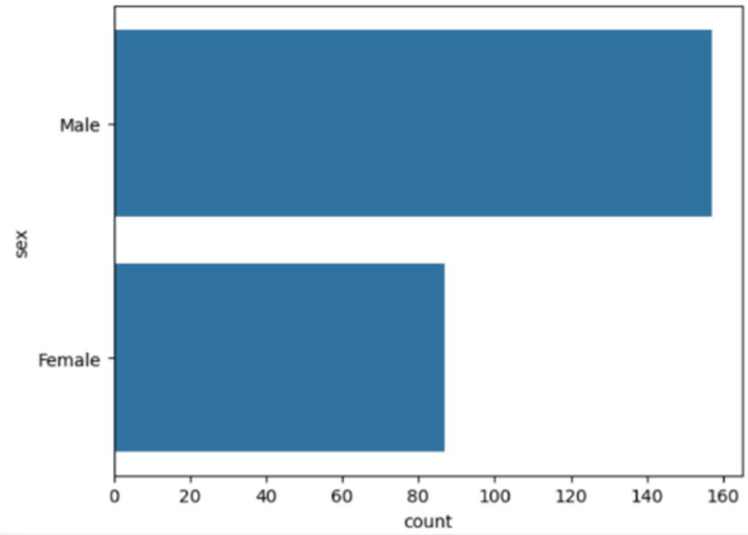

```
sns.countplot(tips.day)
```

```
sns.countplot(tips.day)
```

```
<Axes: xlabel='count', ylabel='day'>
```



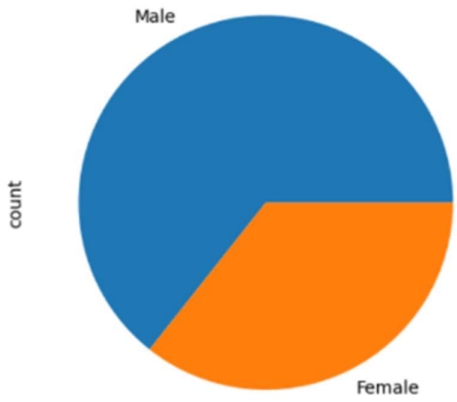

```
<Axes: xlabel='count', ylabel='sex'>
```




```
tips.sex.value_counts().plot(kind='pie')
```

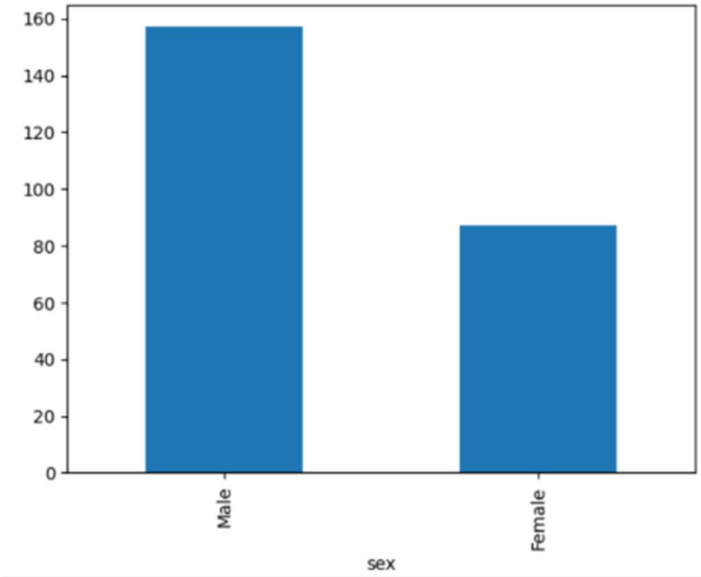
```
tips.sex.value_counts().plot(kind='pie')
```

```
<Axes: ylabel='count'>
```




```
tips.sex.value_counts().plot(kind='bar')
```

```
<Axes: xlabel='sex'>
```




```
sns.countplot(tips[tips.time=='Dinner']['day'])
```

```
<Axes: xlabel='count', ylabel='day'>
```



RESULT:

Thus the EDA qualitative and quantitative analysis experiment is done.

Exp No:17

Experiment to understand Linear Regression for a given data set

AIM:

Understand the Linear regression for the dataset given.

In [ ]:
```
import numpy as np
import pandas as pd
df=pd.read_csv('Salary_data.csv')
df
```

In [19]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64  
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

In [3]:
```
df.dropna(inplace=True)
```

In [4]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 2 columns):
 #   Column           Non-Null Count  Dtype  
---  ------           --------------  -----  
 0   YearsExperience  30 non-null     float64
 1   Salary           30 non-null     int64  
dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

In [5]:
```
df.describe()
```

Out[5]:

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

In [6]:
```
features=df.iloc[:,[0]].values
label=df.iloc[:,[1]].values
```

In [7]:
```
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,label,test_size=0.2,random_st
```

In [20]:
```
from sklearn.linear_model import LinearRegression
model=LinearRegression()
model.fit(x_train,y_train)
```

Out[20]:
```
▾ LinearRegression
LinearRegression()
```

In [21]:
```
model.score(x_train,y_train)
```

Out[21]:
```
0.9603182547438908
```

In [23]:
```
model.score(x_test,y_test)
```

Out[23]:
```
0.9184170849214232
```

In [24]:
```
model.coef_
```

Out[24]:
```
array([[9281.30847068]])
```

In [25]:
```
model.intercept_
```

Out[25]:
```
array([27166.73682891])
```

In [26]:
```
import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
```

In [27]:
```
model=pickle.load(open('SalaryPred.model','rb'))
```

In [28]:
```
yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP)
```

```
Enter Years of Experience: 44
```

In [ ]:
```

```

In [29]:
```
print("Estimated Salary for {} years of experience is {}: " .format(yr_of_exp,Salary)
```

```
Estimated Salary for 44.0 years of experience is [[435544.30953887]]:
```

RESULT:

Experiment to understand Linear Regression for a given data set is done.

Exp No:18

Experiment to understand KNN algorithm for a given data set

AIM:

Understand the KNN algorithm for the dataset given.

In [1]:
```
import numpy as np
import pandas as pd
```

In [2]:
```
df=pd.read_csv('Iris.csv')
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   sepal.length  150 non-null    float64
 1   sepal.width   150 non-null    float64
 2   petal.length  150 non-null    float64
 3   petal.width   150 non-null    float64
 4   variety       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

In [3]:
```
df.variety.value_counts()
```

Out[3]:
```
Setosa        50
Versicolor    50
Virginica     50
Name: variety, dtype: int64
```

In [4]:
```
df.head()
```

Out[4]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

In [5]:
```
features=df.iloc[:,:-1].values
label=df.iloc[:,4].values
```

In [6]:
```
from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
```

In [8]:
```
xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,randc
model_KNN=KNeighborsClassifier(n_neighbors=5)
model_KNN.fit(xtrain,ytrain)
```

Out[8]:
```
KNeighborsClassifier()
```
In a Jupyter environment, please rerun this cell to show the HTML representation or trust the notebook.
On GitHub, the HTML representation is unable to render, please try loading this page with nbviewer.org.

In [9]:
```
print(model_KNN.score(xtrain,ytrain))
print(model_KNN.score(xtest,ytest))
```

```
0.9583333333333334
1.0
```

In [10]:
```
from sklearn.metrics import confusion_matrix
confusion_matrix(label,model_KNN.predict(features))
```

Out[10]:
```
array([[50,  0,  0],
       [ 0, 47,  3],
       [ 0,  2, 48]], dtype=int64)
```

In [11]:
```
from sklearn.metrics import classification_report
print(classification_report(label,model_KNN.predict(features)))
```

```
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        50
  Versicolor       0.96      0.94      0.95        50
   Virginica       0.94      0.96      0.95        50

    accuracy                           0.97       150
   macro avg       0.97      0.97      0.97       150
weighted avg       0.97      0.97      0.97       150
```

In [ ]:
```
```

RESULT:

Experiment to understand KNN algorithm for a given data set is done

LOGISTIC REGRESSION

AIM

To understand the Logistic Regression algorithm by classifying tumours as malignant or benign for a given data set.

PROGRAM

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report

# The sigmoid (logistic) function used by Logistic Regression
z = np.linspace(-8, 8, 200)
plt.plot(z, 1 / (1 + np.exp(-z)))
plt.axhline(0.5, color='red', linestyle='--')
plt.title('Sigmoid function')
plt.xlabel('z')
plt.ylabel('probability')
plt.show()

# Load the data set
data = load_breast_cancer(as_frame=True)
df = data.frame
print("Shape :", df.shape)
print(df.iloc[:, [0, 1, 2, 3, -1]].head())
print("Missing values :", df.isnull().sum().sum())
print(df['target'].value_counts())

features = df.drop('target', axis=1)
label = df['target']
x_train, x_test, y_train, y_test = train_test_split(
    features, label, test_size=0.2, random_state=23)

# Feature scaling
sc = StandardScaler()
x_train = sc.fit_transform(x_train)
x_test = sc.transform(x_test)

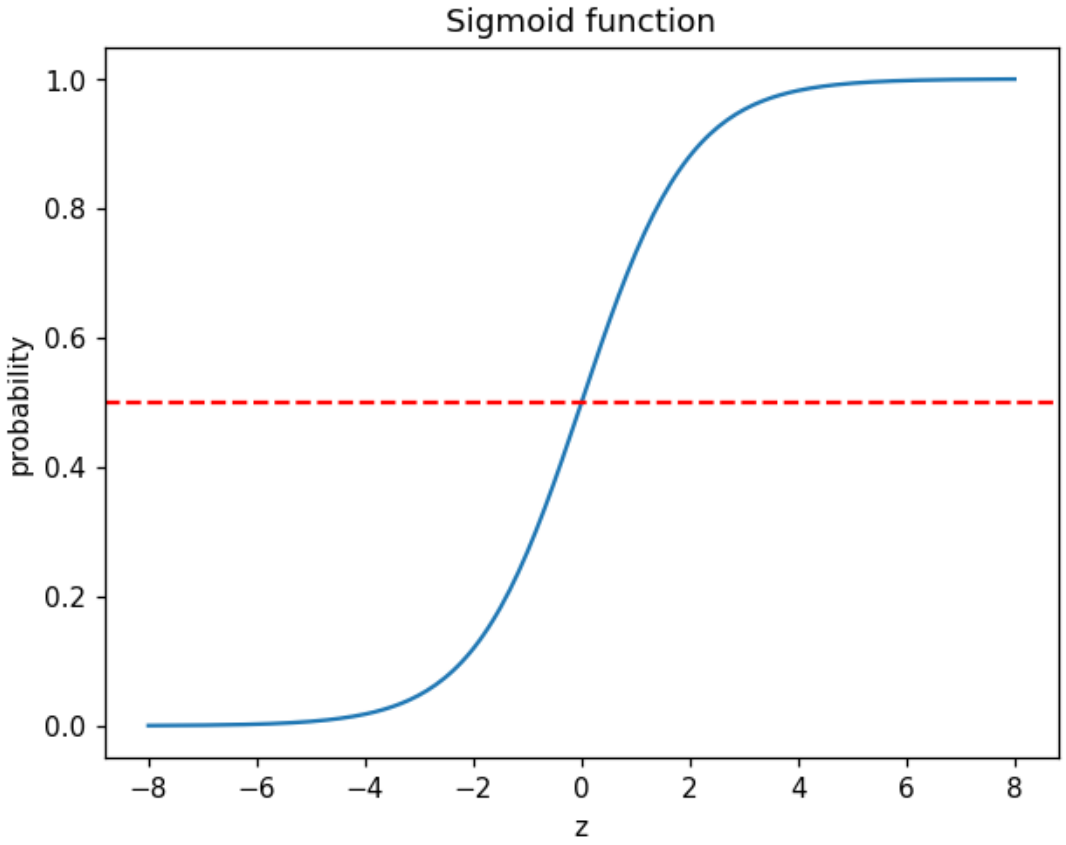
# Training the model
model = LogisticRegression(max_iter=1000)
model.fit(x_train, y_train)

# Prediction and evaluation
y_pred = model.predict(x_test)
print("Training score :", model.score(x_train, y_train))
print("Testing accuracy:", accuracy_score(y_test, y_pred))
print("Probabilities of first 5 test records [malignant, benign]:")
print(model.predict_proba(x_test[:5]).round(4))
print("Predicted classes :", y_pred[:5])
print("Actual classes    :", y_test.values[:5])

cm = confusion_matrix(y_test, y_pred)
print("\nConfusion matrix:\n", cm)
print("\nClassification report:")
print(classification_report(y_test, y_pred, target_names=data.target_names))
```

```
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=data.target_names, yticklabels=data.target_names)
plt.xlabel('Predicted')
plt.ylabel('Actual')
plt.title('Confusion matrix')
plt.show()
```

OUTPUT



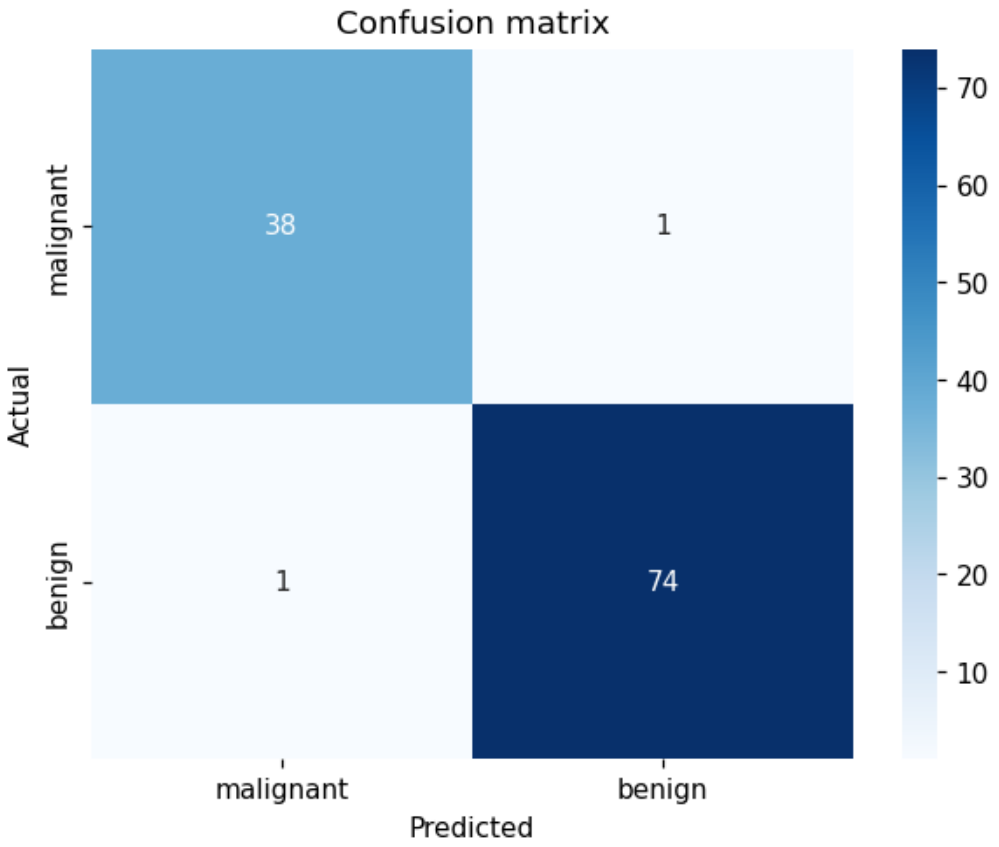

```
Shape : (569, 31)
   mean radius  mean texture  mean perimeter  mean area  target
0        17.99         10.38          122.80     1001.0       0
1        20.57         17.77          132.90     1326.0       0
2        19.69         21.25          130.00     1203.0       0
3        11.42         20.38           77.58      386.1       0
4        20.29         14.34          135.10     1297.0       0
Missing values : 0
target
1    357
0    212
Name: count, dtype: int64
Training score : 0.989010989010989
Testing accuracy: 0.9824561403508771
Probabilities of first 5 test records [malignant, benign]:
[[0.0044 0.9956]
 [0.5369 0.4631]
 [1.     0.    ]
 [0.0336 0.9664]
 [0.9983 0.0017]]
Predicted classes : [1 0 0 1 0]
Actual classes    : [1 1 0 1 0]

Confusion matrix:
 [[38  1]
 [ 1 74]]
```

```
Classification report:
              precision    recall  f1-score   support

   malignant       0.97      0.97      0.97        39
      benign       0.99      0.99      0.99        75

    accuracy                           0.98       114
   macro avg       0.98      0.98      0.98       114
weighted avg       0.98      0.98      0.98       114
```



RESULT

Thus the Logistic Regression model was built and trained on the Breast Cancer data set and it classified the tumours with high accuracy on the test data.

K-MEANS CLUSTERING ALGORITHM

AIM

To understand the K-Means clustering algorithm by grouping the Iris flowers into clusters for a given data set.

PROGRAM

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.metrics import silhouette_score

df = pd.read_csv('Iris.csv')
print(df.head())
features = df.iloc[:, :-1].values        # label is not used for clustering

# Elbow method
wcss = []
for k in range(1, 11):
    km = KMeans(n_clusters=k, init='k-means++', n_init=10, random_state=23)
    km.fit(features)
    wcss.append(km.inertia_)
    print(f"K = {k:<3} WCSS = {km.inertia_:.2f}")

plt.plot(range(1, 11), wcss, marker='o')
plt.title('Elbow method')
plt.xlabel('Number of clusters (K)')
plt.ylabel('WCSS')
plt.show()

# Final model with K = 3
kmeans = KMeans(n_clusters=3, init='k-means++', n_init=10, random_state=23)
labels = kmeans.fit_predict(features)

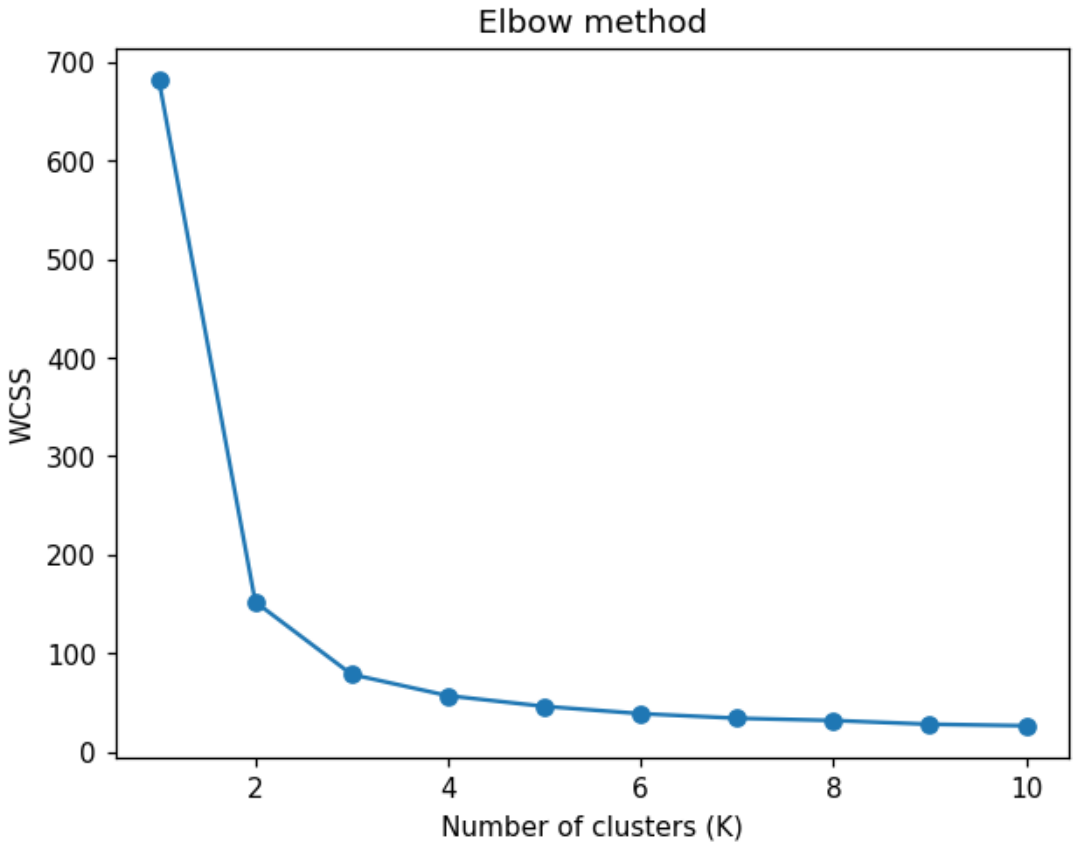
print("\nCluster centres:\n", kmeans.cluster_centers_.round(3))
print("Cluster sizes :", np.bincount(labels))
print("Silhouette score:", round(silhouette_score(features, labels), 4))

# Comparison with the actual species
print("\nClusters vs actual species:")
print(pd.crosstab(df['variety'], labels, colnames=['cluster']))

# Visualization (petal length vs petal width)
fig, ax = plt.subplots(1, 2, figsize=(11, 4))
ax[0].scatter(features[:, 2], features[:, 3], c=labels, cmap='viridis')
ax[0].scatter(kmeans.cluster_centers_[:, 2], kmeans.cluster_centers_[:, 3],
              s=200, c='red', marker='X', label='Centroids')
ax[0].set_title('K-Means clusters')
ax[0].set_xlabel('petal.length')
ax[0].set_ylabel('petal.width')
ax[0].legend()
actual = df['variety'].astype('category').cat.codes
ax[1].scatter(features[:, 2], features[:, 3], c=actual, cmap='viridis')
ax[1].set_title('Actual species')
ax[1].set_xlabel('petal.length')
ax[1].set_ylabel('petal.width')
plt.tight_layout()
plt.show()
```

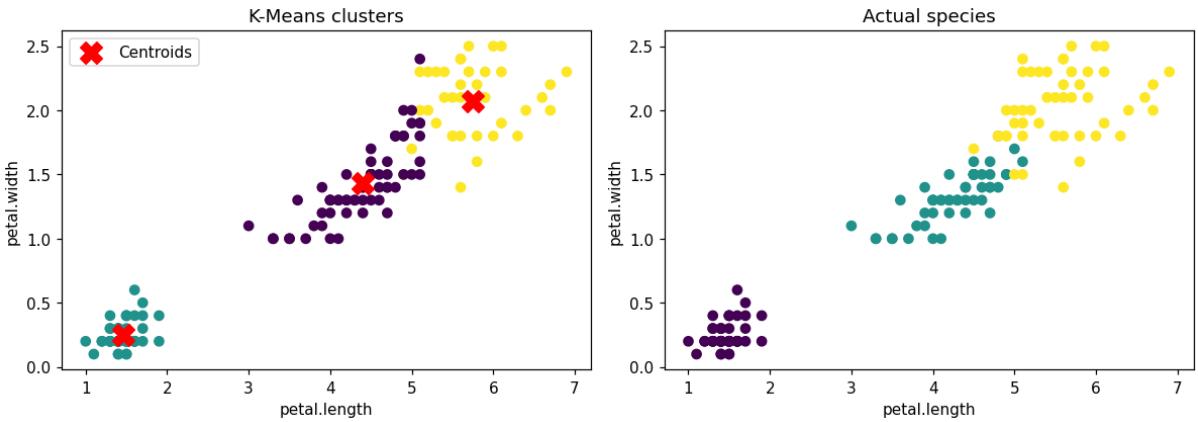
OUTPUT

```
   sepal.length  sepal.width  petal.length  petal.width variety
0           5.1          3.5           1.4          0.2  Setosa
1           4.9          3.0           1.4          0.2  Setosa
2           4.7          3.2           1.3          0.2  Setosa
3           4.6          3.1           1.5          0.2  Setosa
4           5.0          3.6           1.4          0.2  Setosa
K = 1   WCSS = 681.37
K = 2   WCSS = 152.35
K = 3   WCSS = 78.85
K = 4   WCSS = 57.23
K = 5   WCSS = 46.45
K = 6   WCSS = 39.04
K = 7   WCSS = 34.41
K = 8   WCSS = 32.10
K = 9   WCSS = 28.32
K = 10  WCSS = 26.81
```




```
Cluster centres:
 [[5.902 2.748 4.394 1.434]
 [5.006 3.428 1.462 0.246]
 [6.85  3.074 5.742 2.071]]
Cluster sizes : [62 50 38]
Silhouette score: 0.5528

Clusters vs actual species:
cluster      0   1   2
variety
Setosa       0  50   0
Versicolor  48   0   2
Virginica   14   0  36
```



RESULT

Thus the K-Means clustering algorithm was implemented on the Iris data set. The Elbow method gave K = 3 and the clusters closely match the three actual species.

Exp No:1.d

Conduct an experiment to encrypt and decrypt given sensitive data.

Aim:

To conduct an experiment to encrypt and decrypt given sensitive data.

Description:

Use the cryptography library to encrypt and decrypt a piece of data.

Code:

```
# Generate key and encrypt data
from cryptography.fernet import Fernet
key = Fernet.generate_key()
f = Fernet(key)
token = f.encrypt(b"Rajalakshmi Engineering College")
token
b'...'
f.decrypt(token)
b'Rajalakshmi Engineering College'
key = Fernet.generate_key()
cipher_suite = Fernet(key)
plain_text = b"Rajalakshmi Engineering College."
cipher_text = cipher_suite.encrypt(plain_text)
# Decrypt data
decrypted_text = cipher_suite.decrypt(cipher_text)
print("Original Data:", plain_text)
print("Encrypted Data:", cipher_text)
print("Decrypted Data:", decrypted_text)
```

OUTPUT:

Original Data: b'Rajalakshmi Engineering College.'

Encrypted Data: b'gAAAAABmmnHEhy1ZC1-Dox-URNAvYqFFGfJemMb_fKmXqhATe3AAn

XEOnSzaNRINI4zAbNK8lgx2pprrNn-j6agbWDf30OwbrZsBuz8fnDdX7N35WwloDu3m6pVv

kwv8GYDSI8Fov1tV'

Decrypted Data: b'Rajalakshmi Engineering College.'

Result :

Thus an experiment to encrypt and decrypt given sensitive data is conducted.

Exp No:1.c

Conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

Aim:

To conduct an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given.

Description:

Create small datasets for each type and explain their characteristics.

Code:

```
# Structured data example
structured_data = pd.DataFrame({
'ID': [1, 2, 3],
'Name': ['Alice', 'Bob', 'Charlie'],
'Age': [25, 30, 35]
})
print("Structured Data:\n", structured_data)

# Unstructured data example
unstructured_data = "This is an example of unstructured data. It can be a piece of text, an image,
or a video file."
print("\nUnstructured Data:\n", unstructured_data)

# Semi-structured data example (JSON)
semi_structured_data = {'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}}
print("\nSemi-structured Data:\n", semi_structured_data)
```

OUTPUT:

Structured Data:

ID Name Age

0 1 Alice 25

1 2 Bob 30

2 3 Charlie 35

Unstructured Data:

This is an example of unstructured data. It can be a piece of text, an

image, or a video file.

Semi-structured Data:

{'ID': 1, 'Name': 'Alice', 'Attributes': {'Height': 165, 'Weight': 68}

}

Result :

Thus an experiment to differentiate Structured , Un-structured and Semi structured data based on data sets given is conducted.